Aerodynamics and Vehicle Dynamics Course-Work

Module Code: 7037ENG

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1. Introduction

This report analyses both aerodynamic performance and vehicle ride comfort as they play an important role in overall vehicle design. The first part of this report focuses on the aerodynamics of a 20% scale DrivAer model using wind tunnel data together with CFD results. The notchback version of the model is analysed. The study looks at how factors such as rear-end shape, Reynolds number, yaw angle, and underbody design influence drag, lift, and the general flow behaviour around the vehicle.
The second part of the report deals with vehicle dynamics, where a half-car model is used to study ride comfort. MATLAB and Simulink are used to analyse how the vehicle responds to road inputs and how changes in suspension parameters influence performance.

2. Aerodynamics

2.1 Abstract

This study focuses on analysing aerodynamic performance of a 20% scale DrivAer model using both wind tunnel testing and CFD analysis. The main focus was on how rear-end shape, Reynolds number, yaw angle, and underbody design influence the overall airflow behaviour. From the experimental results, the Fastback configuration showed the lowest drag, while the Estate had the highest drag, mainly due to stronger flow separation and a larger wake. The Notchback generally fell in between these two across the different test conditions.

CFD simulations were carried out using a steady-state RANS approach with the SST k-ω model. When compared with the wind tunnel data, the results followed similar trends, especially for lift and pitching moment, although drag was slightly overpredicted. A mesh independence study was also done. The results showed that the chosen mesh was sufficient for reliable predictions.

The study also compared smooth and detailed underbody configurations. The smooth underbody gave lower drag and produced a small amount of downforce, whereas the detailed underbody disturbed the flow more, leading to higher drag, positive lift, and a stronger nose-down pitching effect. Through Flow visualisations more separation and a larger wake for the detailed case are shown.

2.2 Methodology

2.2.1 Wind Tunnel Setup

The wind tunnel balance measures aerodynamic moments at the position of the overhead balance. Therefore, the recorded pitch and roll moments must be resolved to the model centre using the height difference between the balance and the model, which is 1.17 m.

For the Notchback model at approximately 0° yaw and 24.98 m/s, the results give:

- Drag force, $D = 8.4435$ N
- Side force, $Y = 2.1437$ N
- Recorded pitch moment, $M_p = -12.2586$ Nm
- Recorded roll moment, $M_r = 2.6861$ Nm

The resolved pitch moment at the model centre is:

$$M_{p,res} = M_p + D \times h = -2.3797 \text{ Nm}$$

This agrees with the dataset value of approximately -2.3796 Nm.

The resolved roll moment at the model centre is:

$$M_{r,res} = M_r - Y \times h = 0.1780 \text{ Nm}$$

This matches the result of approximately 0.1779 Nm.

For the aerodynamic coefficients, the reference frontal area is 0.087 m², and the wheelbase used as the reference length for moment coefficient is 0.56 m.

The drag coefficient is calculated as:

$$C_D = \frac{F_x}{0.5\rho V^2 A}$$

The lift coefficient is calculated as:

$$C_L = \frac{F_z}{0.5\rho V^2 A}$$

The pitching moment coefficient is calculated as:

$$C_M = \frac{M_{p,res}}{0.5\rho V^2 A L}$$

Using the same 0° yaw condition:

- $\rho = 1.1888$ kg/m^3
- $V = 24.981$ m/s
- $A = 0.087$ m^2
- $L = 0.56$ m

- $F_x = 8.4436$ N
- $F_z = 1.9289$ N
- $M_{p,res} = -2.3796$ Nm

Dynamic pressure term:

$0.5\rho V^2 A \approx 32.27$

So,

$$C_D = \frac{8.4436}{32.27} \approx 0.262$$

$$C_L = \frac{1.9289}{32.27} \approx 0.060$$

$$C_M = \frac{-2.3796}{32.27 \times 0.56} \approx -0.132$$

Same equations are used to derive and understand the results for the other models also.

2.2.3 CFD Modelling approach

2.2.3.1 Vehicle Dimensions

The model vehicle dimensions used in the simulation were:

Length, $L = 0.92460\ m$,Width, $W = 0.33956\ m$, Height, $H = 0.26605\ m$

These dimensions were used to determine the wind tunnel domain size, blockage ratio, Reynolds number and the aerodynamic reference conditions

Frontal Area of the Vehicle

The frontal area was calculated using the vehicle width and height: $A_f = W \times H$

Substituting the model dimensions:$A_f = 0.33956 \times 0.26605 = 0.09034\ m^2$

Therefore, the frontal area of the vehicle model is: $A_f = 0.09034\ m^2$

This value was later used in the blockage ratio assessment and in aerodynamic coefficient calculations.

2.2.3.2 Wind Tunnel Length Determination

The computational wind tunnel was sized to provide sufficient space for flow development both upstream and downstream of the vehicle. A minimum distance of two vehicle lengths in front and five vehicle lengths behind the model was adopted.

Upstream distance

$$L_{front} = 2L = 2 \times 0.92460 = 1.84920\ m$$

Downstream distance

$$L_{back} = 5L = 5 \times 0.92460 = 4.62300\ m$$

Total tunnel length

The total tunnel length was then obtained by adding the front clearance, vehicle length and rear clearance:

$L_{tunnel} = L_{front} + L + L_{back} = 7.39680\ m$

Therefore, the selected wind tunnel length was:

$L_{tunnel} = 7.39680\ m$

This arrangement provides adequate space for the inflow to stabilise before reaching the vehicle and allows the wake to develop sufficiently behind it.

Blockage Ratio Assessment

The blockage ratio is used to evaluate whether the computational tunnel cross-section is sufficiently large relative to the vehicle frontal area. It is defined as:

Blockage Ratio $= \frac{A_f}{A_t} \times 100$

where:

- A_f= vehicle frontal area
- A_t= tunnel cross-sectional area

For a target blockage ratio of 1%, the minimum required tunnel area is:

$$A_t = \frac{A_f}{0.01}$$

Substituting the frontal area:

$$A_t = \frac{0.09034}{0.01} = 9.034\ m^2$$

Therefore, a tunnel cross-sectional area greater than 9.034 $m^2$is required to keep the blockage ratio at or below 1%.

Rather than using the absolute minimum, a larger cross-section was selected to reduce wall interference further. A tunnel width of 3.2 m and height of 3.0 m were adopted.

$$A_t = 3.2 \times 3.0 = 9.6\ m^2$$

The resulting blockage ratio becomes:

$$\frac{0.09034}{9.6} \times 100 = 0.941\%$$

Hence, the final blockage ratio is $= 0.94\%$

This confirms that the selected domain satisfies the requirement of remaining below 1% blockage.

Reynolds Number Calculation

The Reynolds number was calculated to determine the flow regime around the vehicle. The standard equation is:

$$Re = \frac{\rho V L}{\mu}$$

where:

- ρ= air density
- V= flow velocity
- L= characteristic length
- μ= dynamic viscosity of air

For the present simulation:

$\rho = 1.225\ kg/m^3$

$V = 45\ m/s$

$L = 0.92460\ m$

$\mu = 1.81 \times 10^{-5}\ kg/(m \cdot s)$

Substituting these values:

$$Re = \frac{1.225 \times 45 \times 0.92460}{1.81 \times 10^{-5}} = \frac{50.973525}{1.81 \times 10^{-5}} \approx 2.82 \times 10^6$$

This indicates the turbulent flow, which justifies the use of a turbulence model in the simulation.

2.2.3.3 Near-Wall Mesh Estimation Using y^+

To estimate a suitable first cell height near the vehicle surface, the wall y^+relationship was used. This helps in selecting the prism layer thickness.

The first equation required is an estimate of the skin friction coefficient:

$$C_f = 0.058Re^{-0.2}$$

Using:

$Re = 2.82 \times 10^6$

$C_f = 0.058(2.82 \times 10^6)^{-0.2} = 0.00297$

The wall shear stress is then estimated by:

$\tau_w = \frac{1}{2} C_f \rho U^2$

$\tau_w = \frac{1}{2}(0.00297)(1.225)(45^2) \approx 3.69\ N/m^2$

The friction velocity is then calculated as:

$$U_\tau = \sqrt{\frac{\tau_w}{\rho}} = \sqrt{\frac{3.69}{1.225}} = \sqrt{3.012} \approx 1.73\ m/s$$

Finally, the first cell height is estimated from:

$y = \frac{y^+ \mu}{\rho U_\tau}$

For $y^+ = 50$

the first cell height becomes:

$$y = \frac{50 \times 1.81 \times 10^{-5}}{1.225 \times 1.73} = \frac{9.05 \times 10^{-4}}{2.11925} = 4.27 \times 10^{-4}\ m \approx 0.43\ mm$$

Therefore, the estimated first near-wall cell height for a target $y^+$of 50 is $\approx 0.43\ mm$

Prism Layer Mesh Configuration

Due to computational limitations, the prism layer settings were simplified to:

Total prism layer thickness (T) = 3.5 mm

Number of prism layers (a) = 3

Stretching factor (r) = 1.3

To determine the first prism layer thickness, the geometric series relationship was used:

$$T = a(1 + r + r^2)$$

$$3.5 = a(1 + 1.3 + 1.69)$$

$$3.5 = a(3.99)$$

$$a = \frac{3.5}{3.99} = 0.877 \text{ mm}$$

Thus, the first prism layer thickness is approximately:

$$a \approx 0.88\ mm$$

It indicates that the selected prism mesh remains acceptable as a practical compromise.

Category	Details
Governing Equations	Reynolds-Averaged Navier–Stokes (RANS) equations used to model fully turbulent flow and capture mean aerodynamic behaviour efficiently.
Turbulence Model	SST k-ω model selected due to its ability to predict boundary layer behaviour, adverse pressure gradients, and flow separation, especially in rear and underbody regions.
Flow Assumptions	Flow assumed to be steady and incompressible, as the inlet velocity (45 m/s) is below the compressibility threshold.
Wall Treatment	All-high y^+ wall treatment used to accommodate the mesh resolution and provide flexibility for near-wall modelling with moderate prism layers.
Boundary Conditions	Slip wall conditions applied to wind tunnel boundaries to eliminate wall shear effects and simulate an open flow environment.
Blockage Ratio	Domain size selected to maintain a low blockage ratio (<1%), reducing wall interference effects.
Coordinate System	Cartesian coordinate system used for moment calculations.
Reference Origin	Origin set at (0, 0, 0.3 m).
Axis Directions	X – longitudinal direction, Y – lateral direction, Z – vertical direction.
Moment Calculation	Pitching moment evaluated about the Y-axis (lateral axis).

Table 1: Physics Model and Boundary Setup (refer appendix A)

2.3 Wind tunnel data results

The wind tunnel tests data included measurements of the aerodynamic forces, and pitching moment, and the readings of the pressure at multiple taps across the model. The measurements were then taken at different Reynolds numbers and yaw angles as given in the readings to capture how the flow behaved under various conditions.

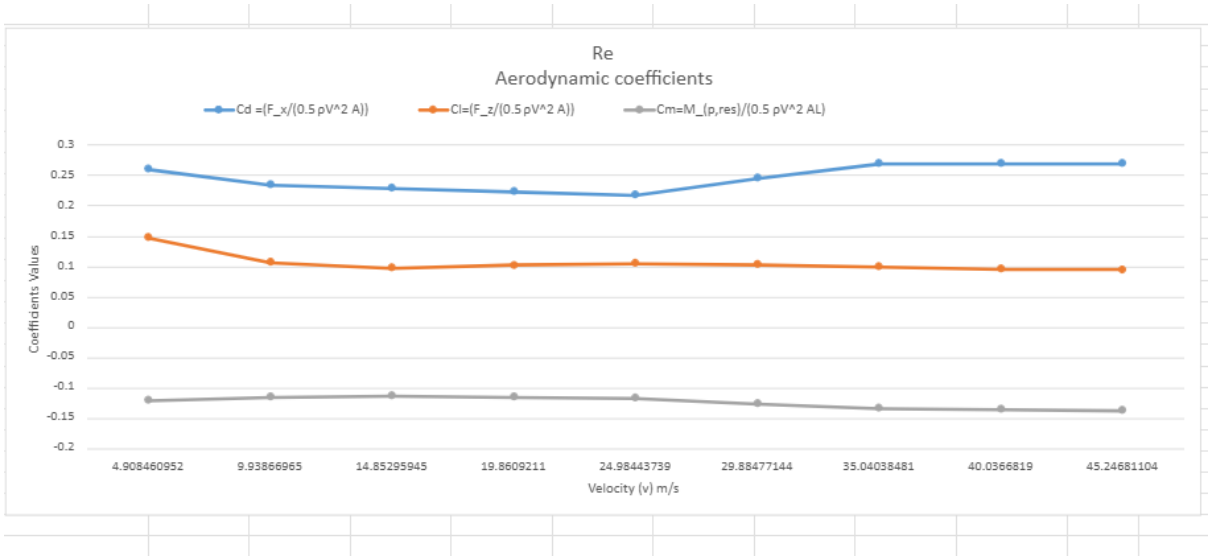


Figure 1: Aerodynamics coefficient vs velocity

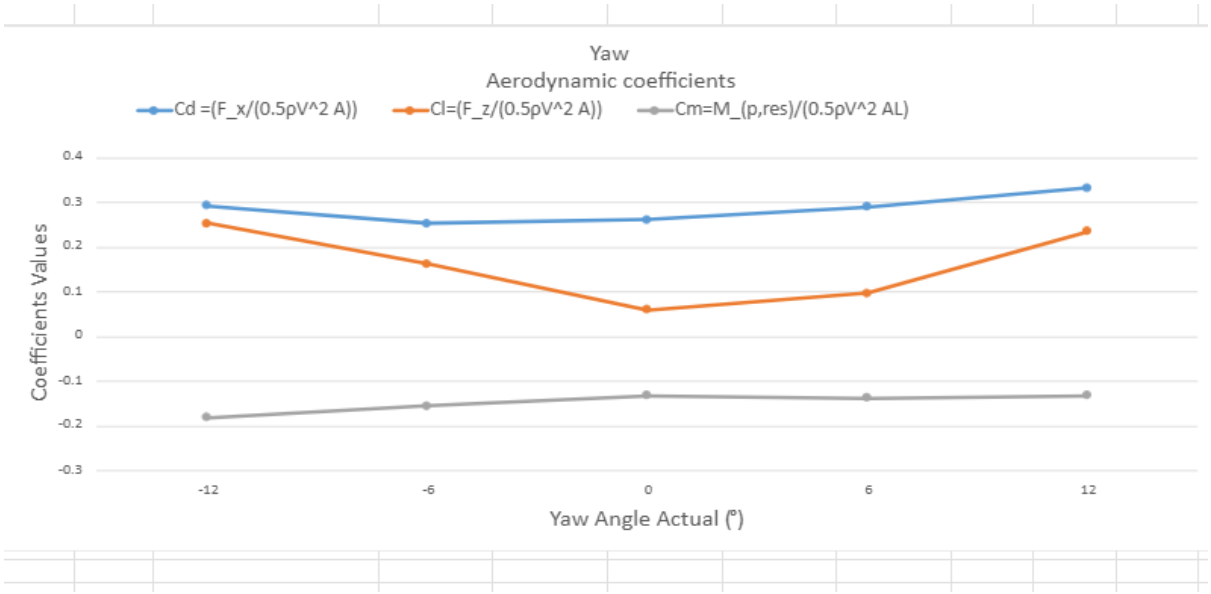


Figure 2: Aerodynamics coefficient vs Yaw angle

Reynolds Number Sweep Analysis

To investigate the influence of flow velocity on aerodynamic performance, a Reynolds number sweep was conducted for all three rear-end configurations: fastback, notchback, and estate.

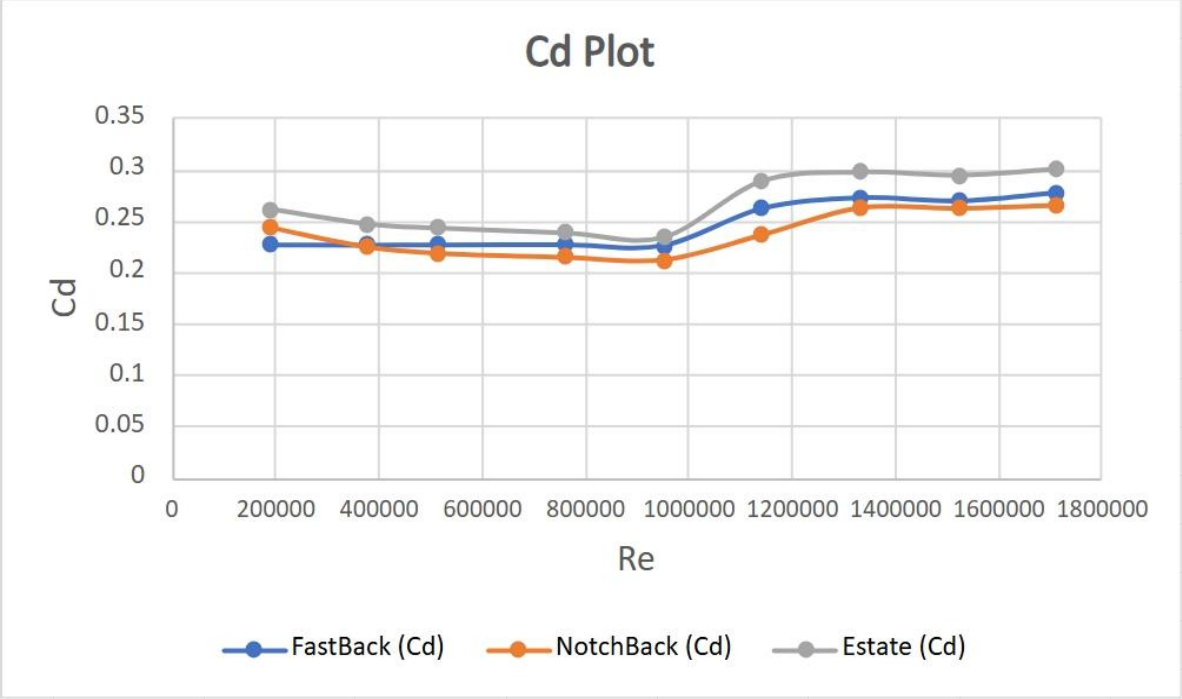


Figure 3: CD vs Reynolds Number

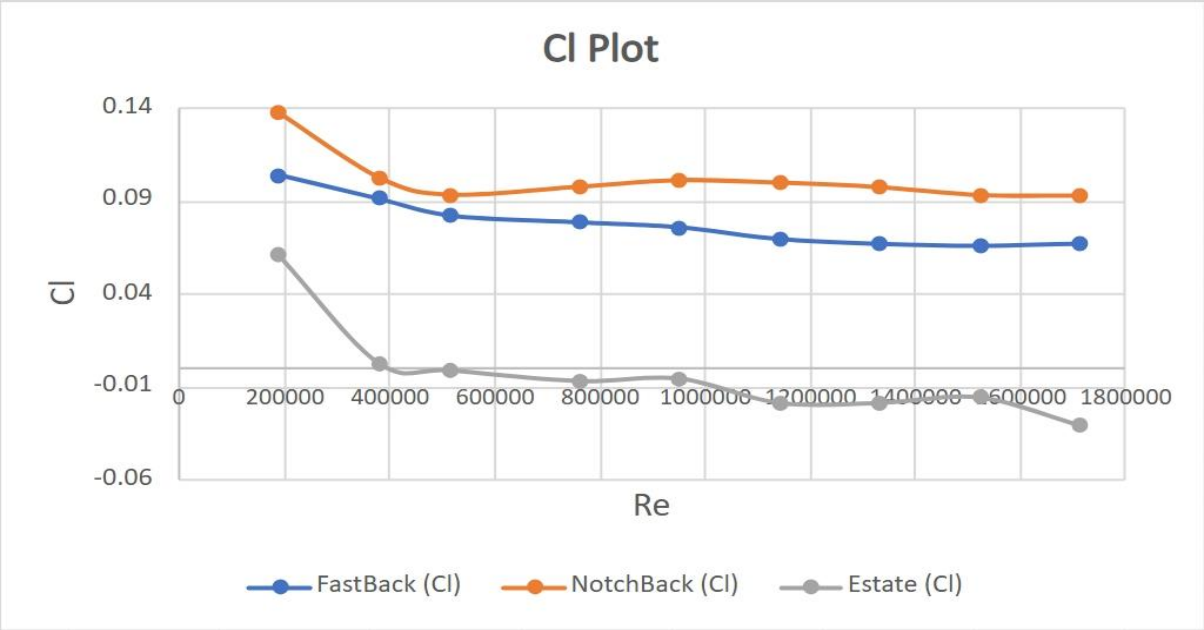


Figure 4: CL vs Reynolds Number

Yaw Angle Analysis

To assess the effect of crosswind conditions, aerodynamic coefficients were evaluated at a constant wind speed of 25 m/s for yaw angles of -12°, -6°, 0°, 6°, and 12°.

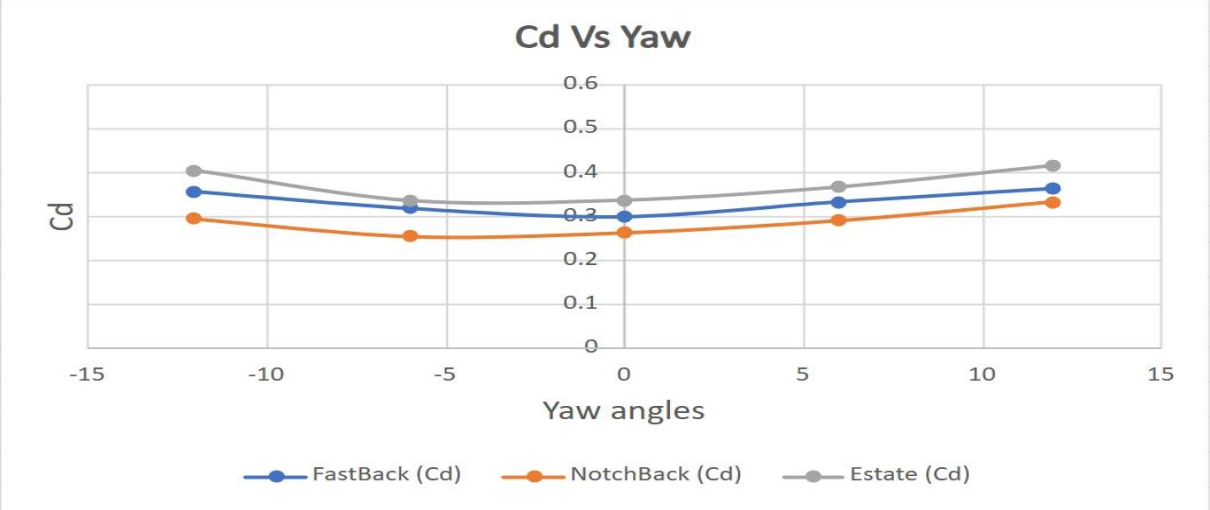


Figure 5: CD vs Yaw Angle

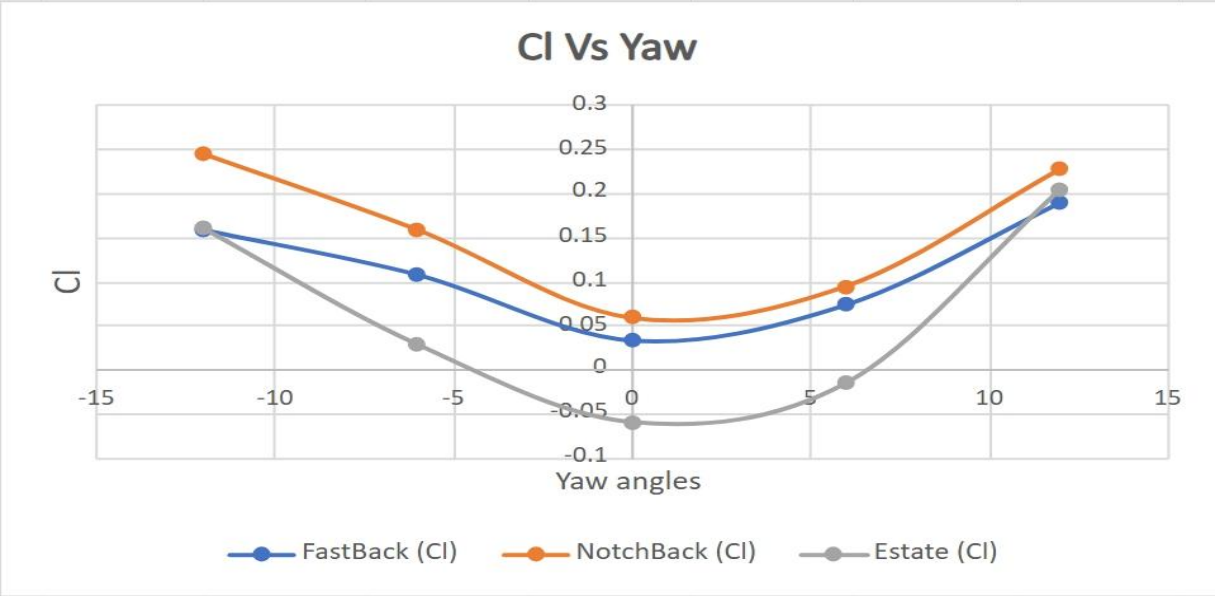


Figure 6: CL vs Yaw Angle

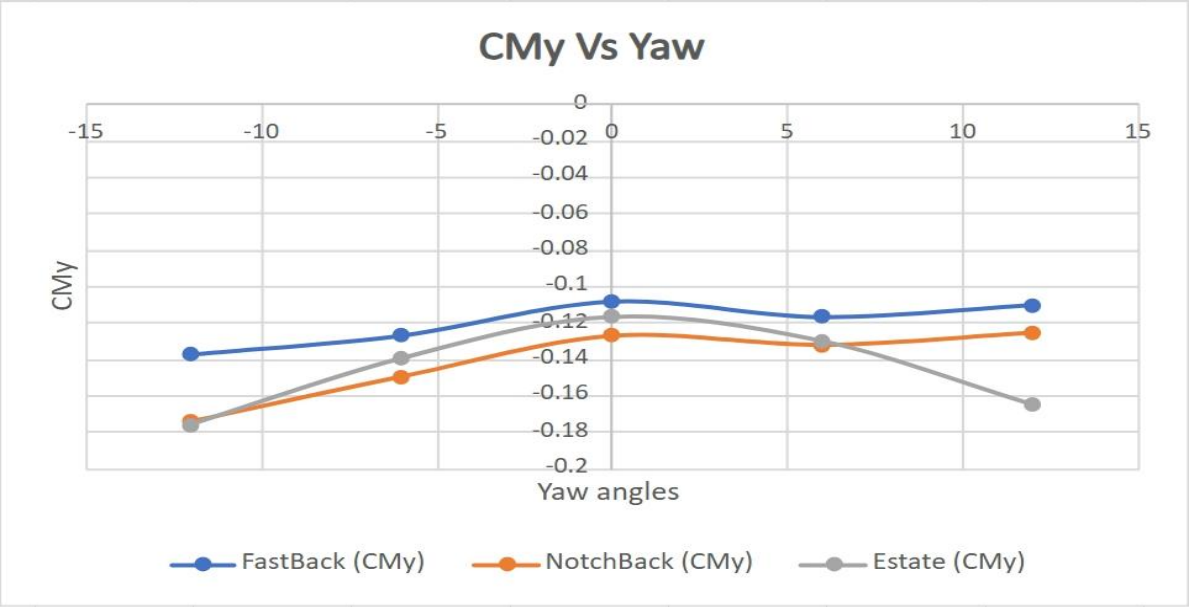


Figure 7: CMy vs Yaw Angle

Pressure Coefficient Distribution

The pressure coefficient distribution was calculated to investigate flow behaviours around the vehicle.

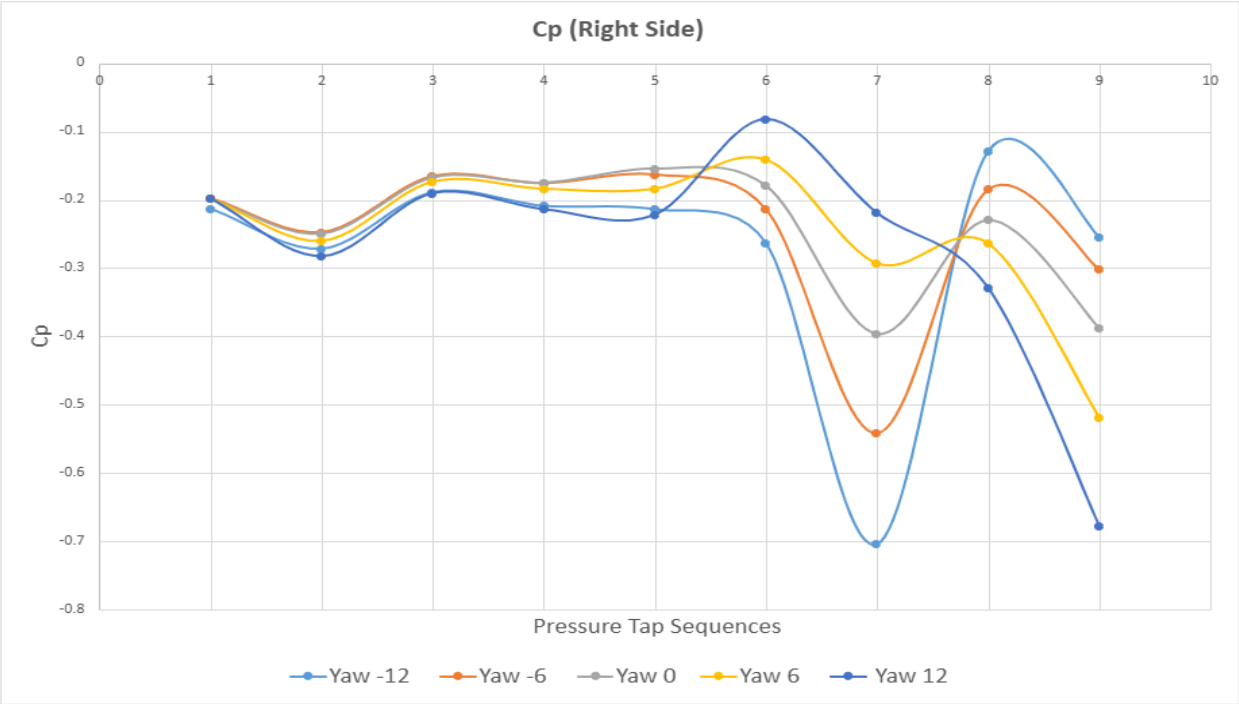


Figure 8: Cp distribution - Right Side

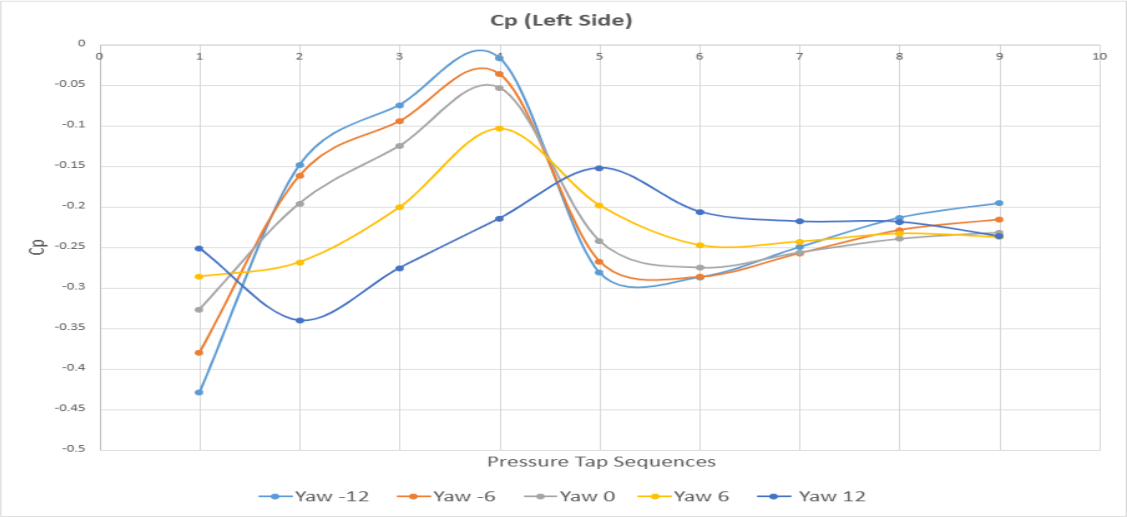


Figure 9: Cp distribution - Left Side

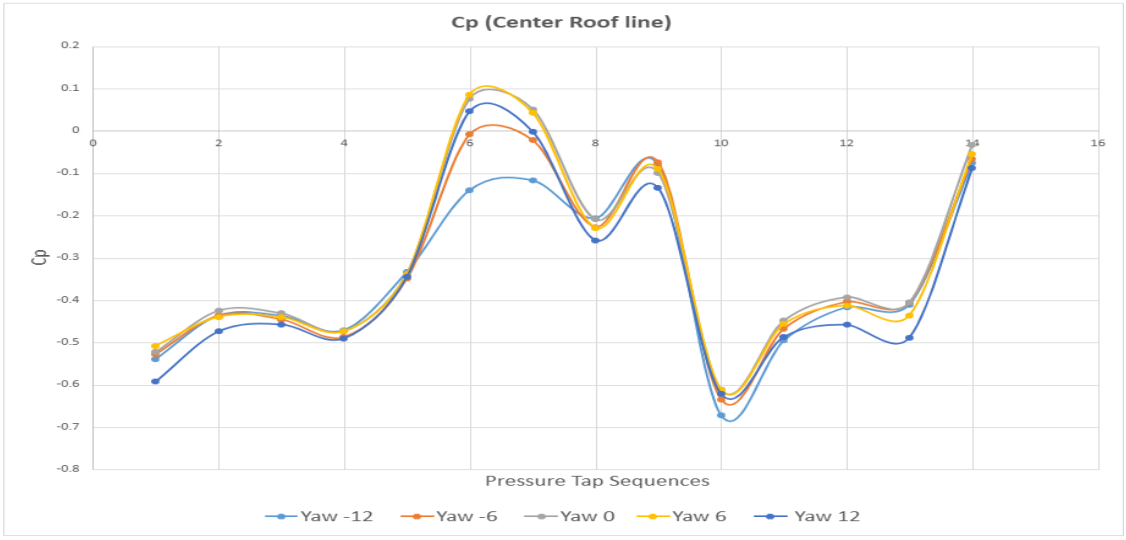


Figure 10: Cp distribution - Centre Roof Line

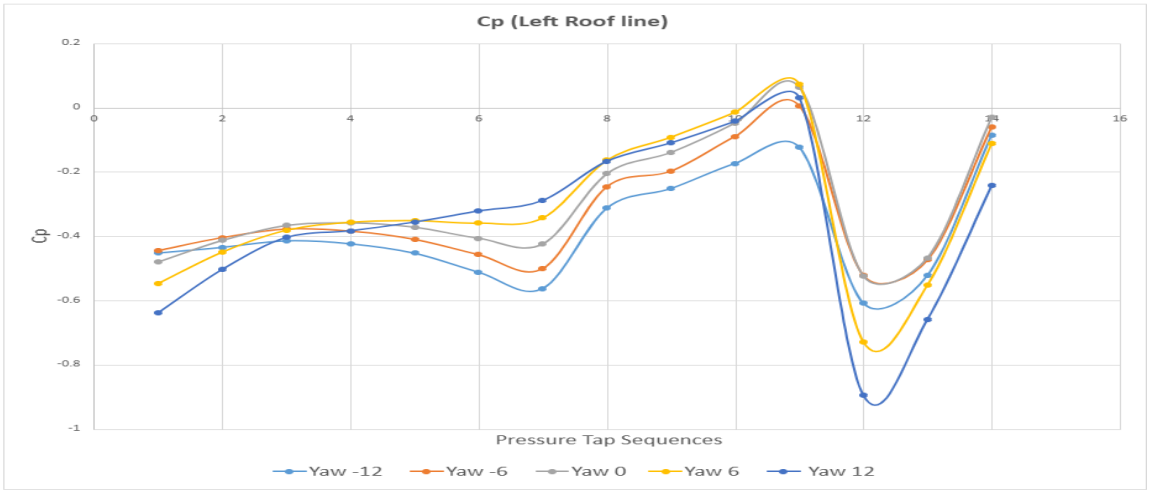


Figure 11: Cp distribution - Left Roof Line

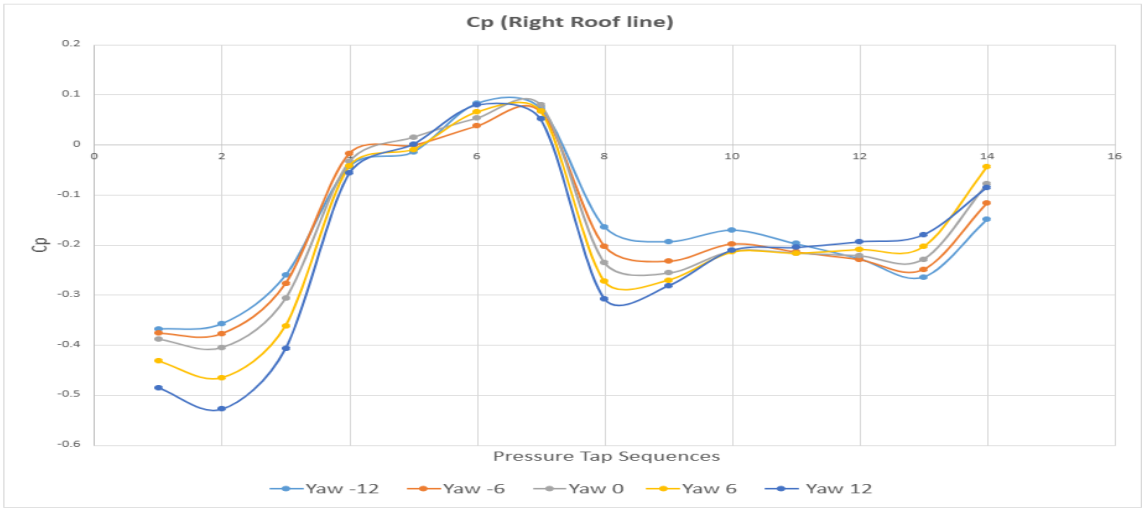


Figure 12: Cp distribution - Right Roof Line

2.4. Wind tunnel data discussion

2.4.1 Reynolds Number Sweep Results (refer appendix A for detailed explaination)

Parameter	Fastback	Notchback	Estate
Drag Coefficient (Cd)	Lowest and stable (≈ 0.23–0.26); indicates delayed separation and good pressure recovery (Hucho, 2013; Katz, 2016)	Intermediate behaviour; slight reduction at mid-range, increases at higher speeds due to transition to wake-dominated flow	Highest drag (≈ 0.30 at high Re); early separation and large wake (Heft et al., 2012; Ahmed et al., 1984)
Lift Coefficient (Cl)	Low and stable values	Higher lift values; slightly decreases with increasing Reynolds number	Decreases towards near-zero or negative at high speeds; linked to strong rear separation (Hucho, 2013; Bearman, 1984)
Pitching Moment (CMy)	Negative and relatively stable (mild nose-down tendency)	Negative; becomes more negative with increasing Reynolds number	Negative with largest variation; influenced by unstable wake effects (Heft et al., 2012; Katz, 2016)
Flow Behaviour	Delayed separation, smaller wake	Partial attachment leads to separation at higher speeds	Early separation, large and unstable wake

Table 2: Aerodynamics coefficients

2.4.2 Yaw Angle Sweep Results

From the Cd vs yaw results it can be identified that drag is lowest at 0° yaw an it increases with yaw due to flow asymmetry and wake growth (Hucho, 2013; Katz, 2016). The Estate model is showing the highest drag (exceeding ≈ 0.40). The Fastback shows the least variation while the Notchback remains intermediate.

The Cl results show increasing lift with yaw an because of asymmetric pressure distribution. And estate model is showing the largest variation.

The CMy (pitch moment co efficient) remains negative across all yaw angles,while the Estate model is showing the greatest fluctuations. The Fastback is the most stable and the Notchback shows moderate variation.

2.5 CFD Validation and results

2.5.1 Mesh independency study

Changing the Base size

Mesh Type	Cells	Cd	Cl	Cmy
Coarse	0.5million	0.300	0.052	-0.115
Medium	1.2million	0.299	0.062	-0.129
Fine	2.3million	0.30452	0.09302	-0.14192

Table 3: Base size

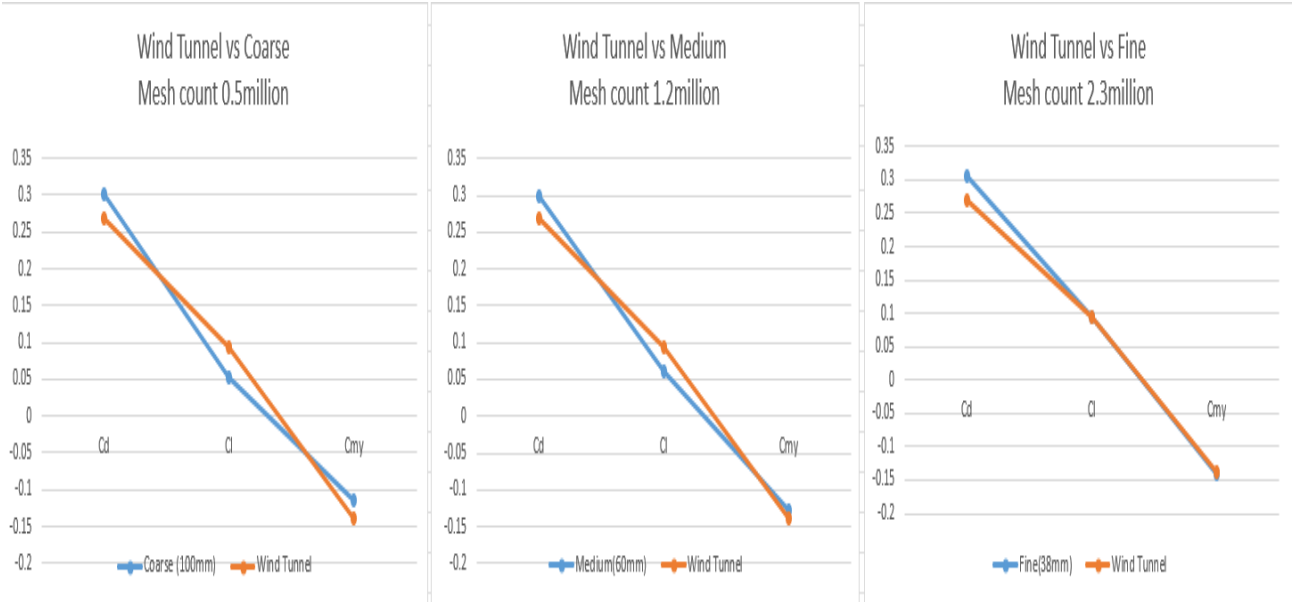


Figure 13: Mesh Independence stud

The mesh independence study shows that the aerodynamic coefficients stabilise as the mesh is refined. From coarse to medium mesh, noticeable changes are observed in Cl (0.052 to 0.062) and CMy (-0.115 to -0.129), indicating improved resolution of flow features. However, from medium to fine mesh, the changes are smaller for Cd (0.299 to 0.3045) but more pronounced for Cl and CMy, suggesting increased sensitivity of lift and pitching moment to mesh refinement. When compared with wind tunnel results, the fine mesh shows the closest overall agreement, particularly in capturing the trends of aerodynamic coefficients. This indicates that the solution is approaching mesh independence, and the fine mesh is sufficiently accurate for further analysis.

CFD results for detailed body

Drag Coefficient (Cd)

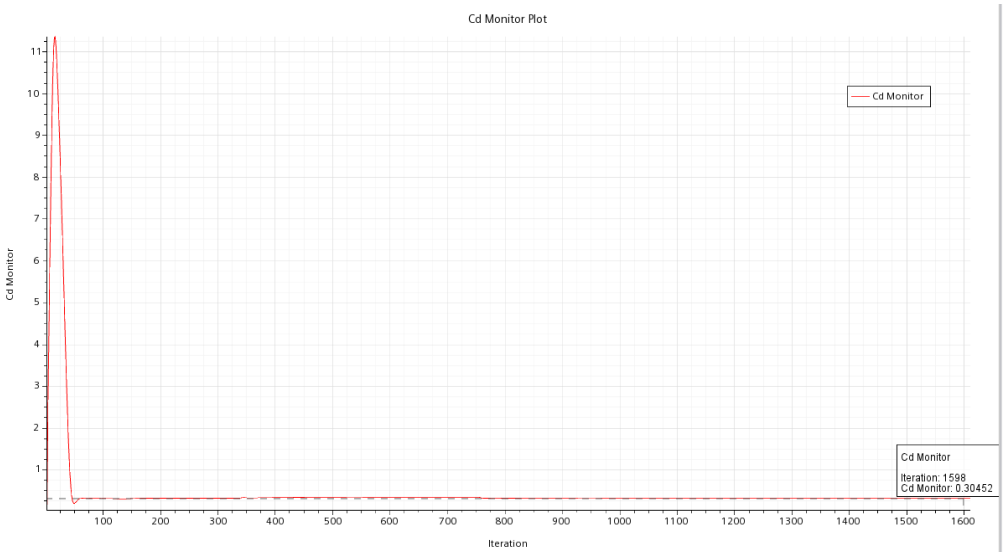


Figure 14: C_d monitor plot (Detailed body)

Lift Coefficient (Cl)

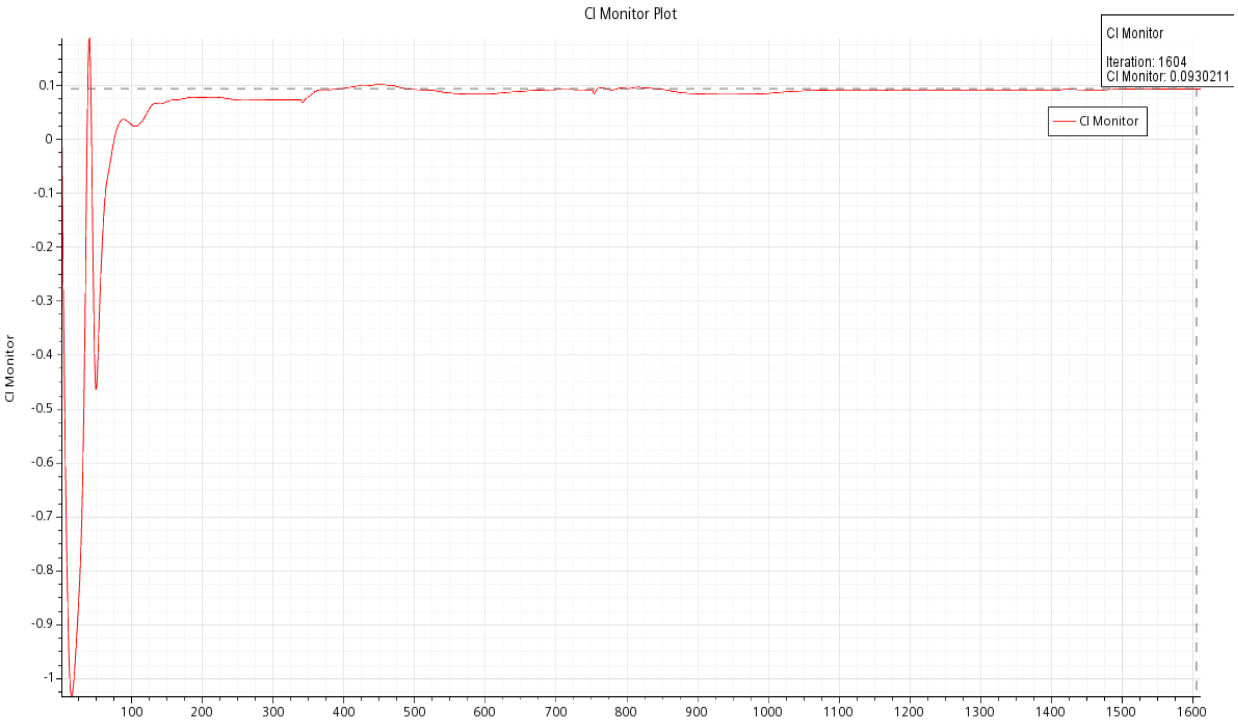


Figure 15: C_l monitor plot (Detailed body)

Pitching Moment Coefficient (CMy)

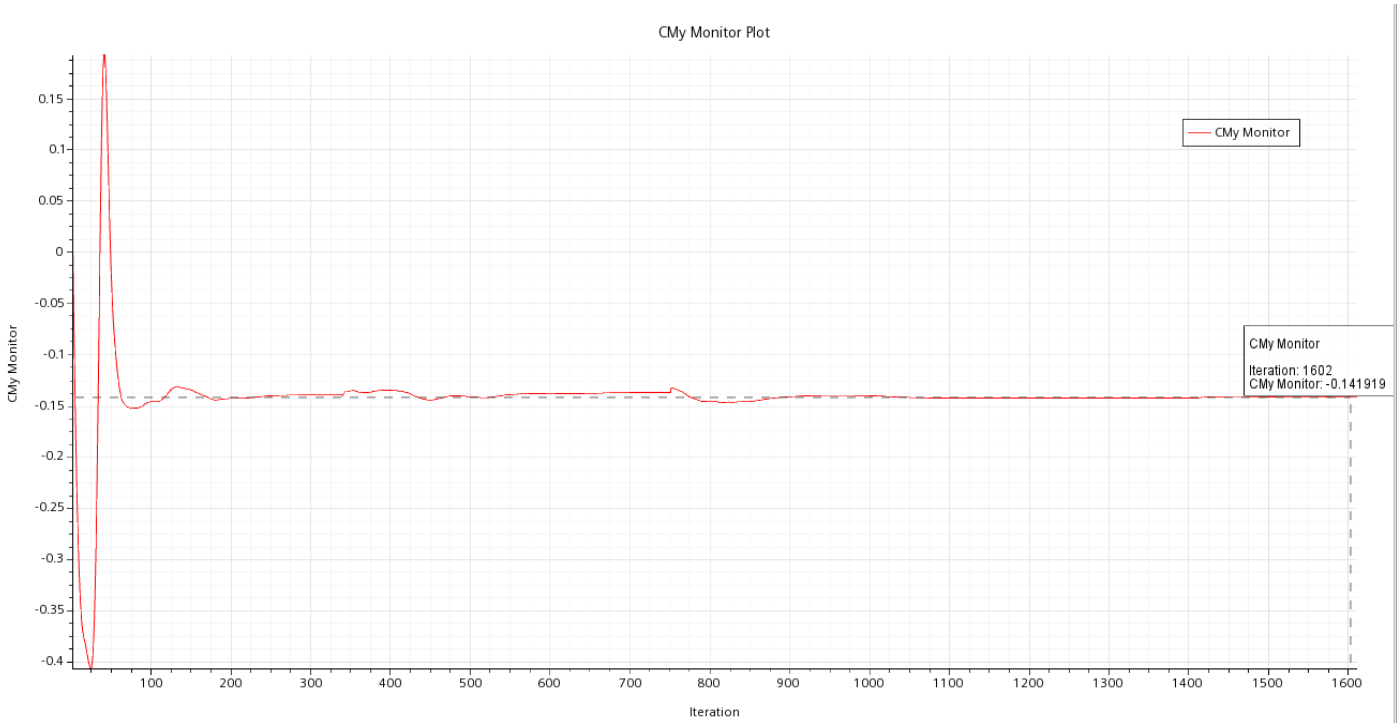


Figure 16: CM_y monitor plot (Detailed body)

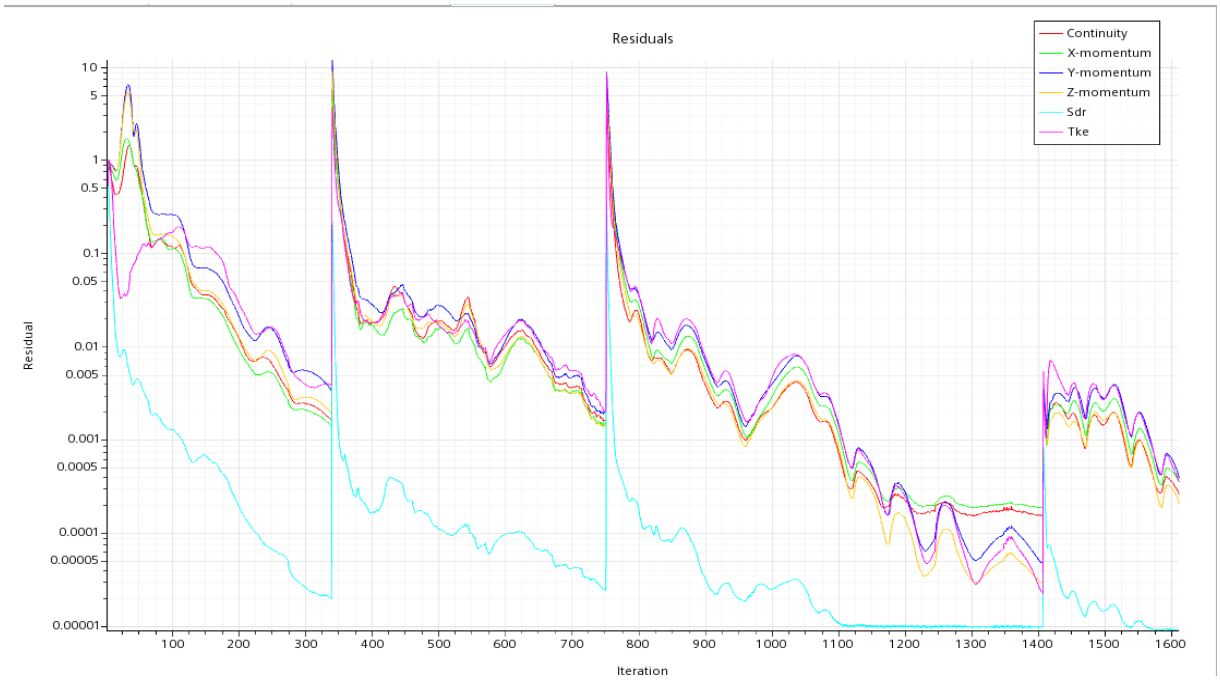


Figure 17: Residuals plot (Detailed body)

CFD results for smooth body

Lift Coefficient (Cl)

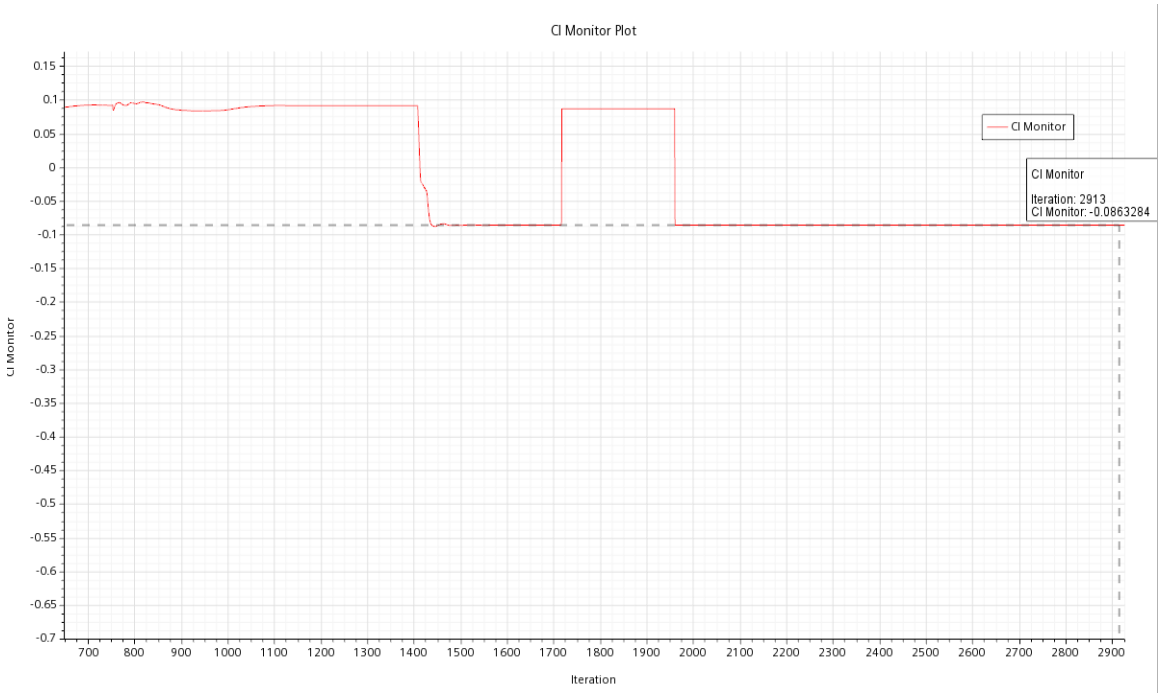


Figure 18: C_l monitor plot (Smooth body)

Drag Coefficient (Cd)

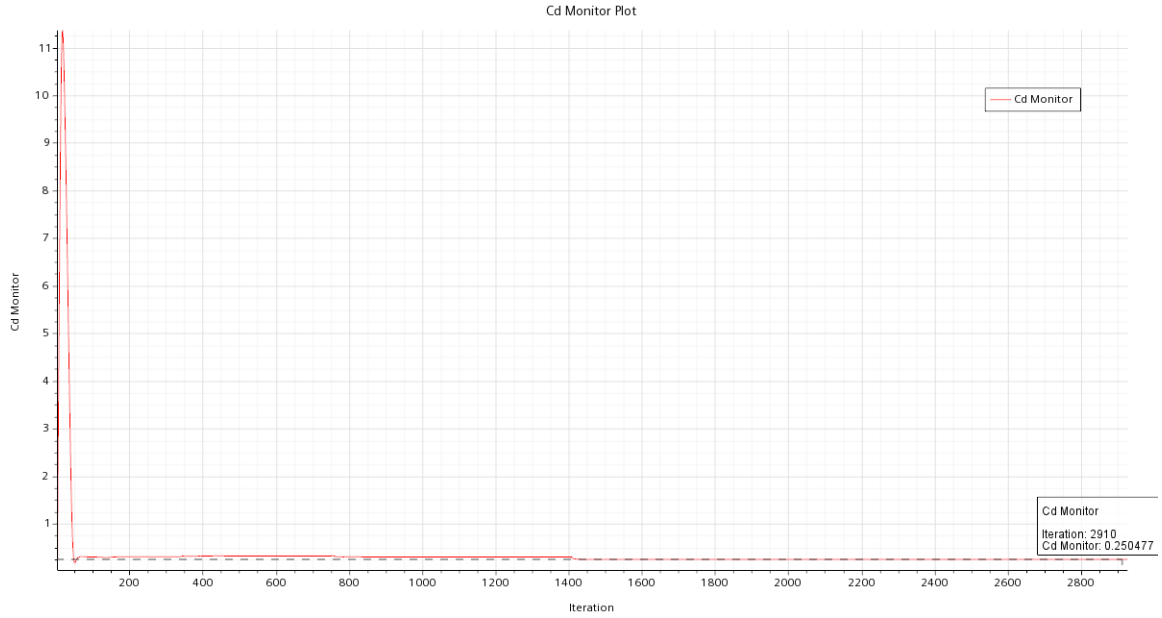


Figure 19: C_d monitor plot (Smooth body)

Pitching Moment Coefficient (CMy)

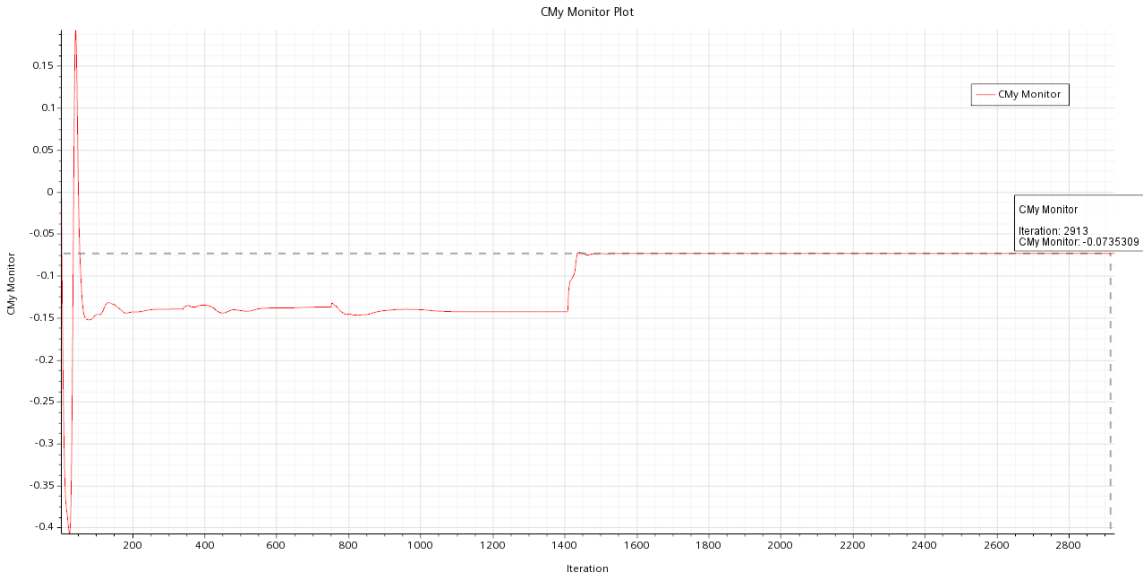


Figure 20: CM_y monitor plot (Smooth body)

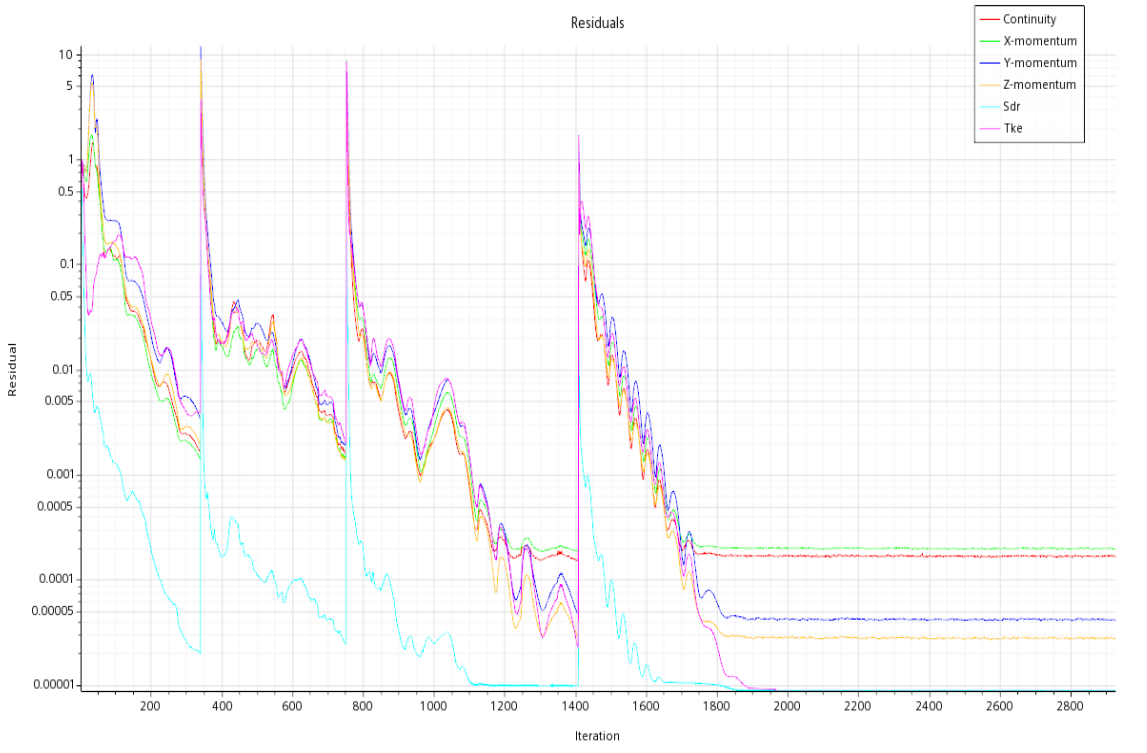


Figure 21: Residuals (Smooth body)

Comparing experimental data with the numerical results

For the Notchback configuration with a detailed underbody at a wind speed of 45 m/s and 0° yaw, the CFD results are taken from STAR-CCM+ are shown in table 2.

CFD result shows that the numerical model is producing a larger wake or stronger flow separation, particularly around the rear and underbody regions.

For the lift coefficient, the CFD prediction has a very close agreement with the wind tunnel value. This indicates that the simulation has been able to capture the overall pressure distribution across the vehicle body effectively.

The pitching moment coefficient shows a small deviation. It indicates a marginally stronger nose-down moment which was predicted by the simulation. This shows that the aerodynamic loading is shifting towards the rear in the CFD model. It is caused by underbody flow behaviour and wake structure differences.

Parameter	Wind Tunnel	CFD (STAR-CCM+)
Cd	0.2697	0.30452
Cl	0.0946	0.09302
CMy	-0.1381	-0.14192

Table 4: Comparison of Wind tunnel data and Star-CMM data

Comparing pressure coefficient distribution

Since the provided pressure data in the wind tunnel are taken when the velocity is 25m/s the simulation was run again for 25 m/s velocity to plot the following plots in order to make a valid argument.

Right side

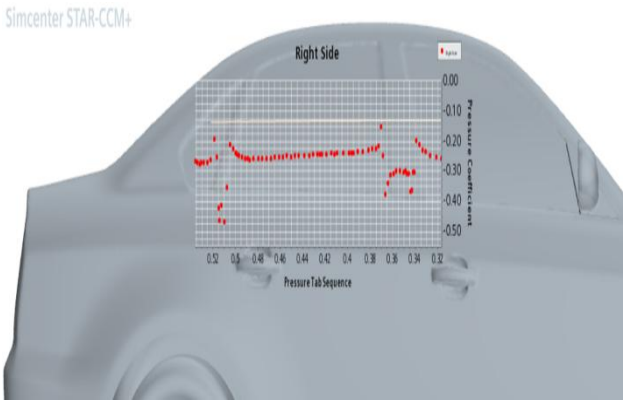


Figure 22: Star CCM result

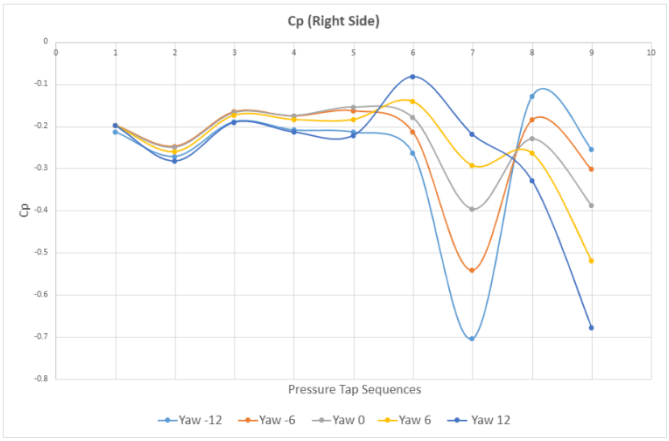


Figure 23: wind tunnel test results

Left side

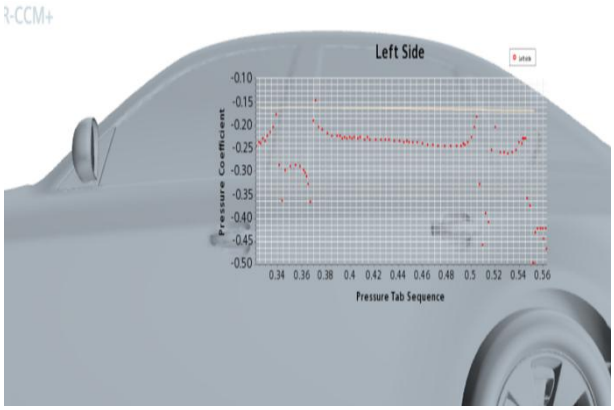


Figure 24: Star CCM + result

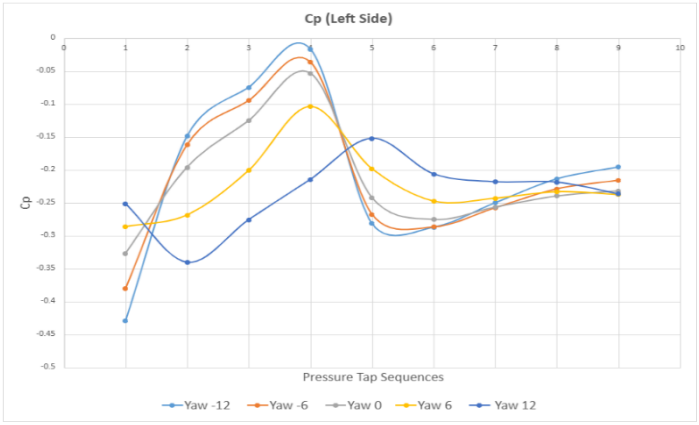


Figure 25: Wind Tunnel test results

Centre Roof line

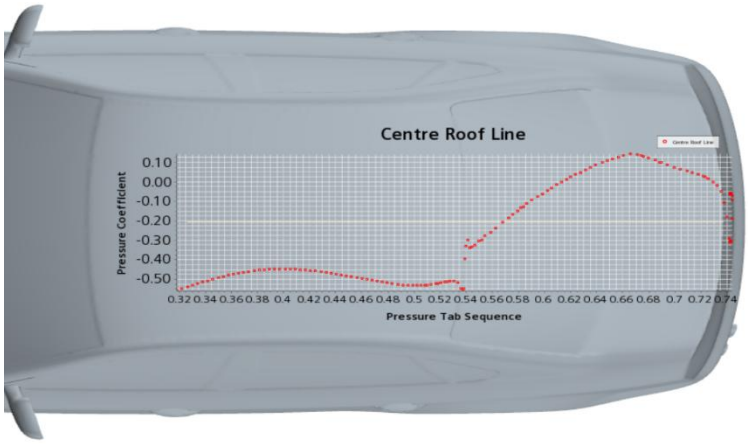


Figure 26: Star CCM result

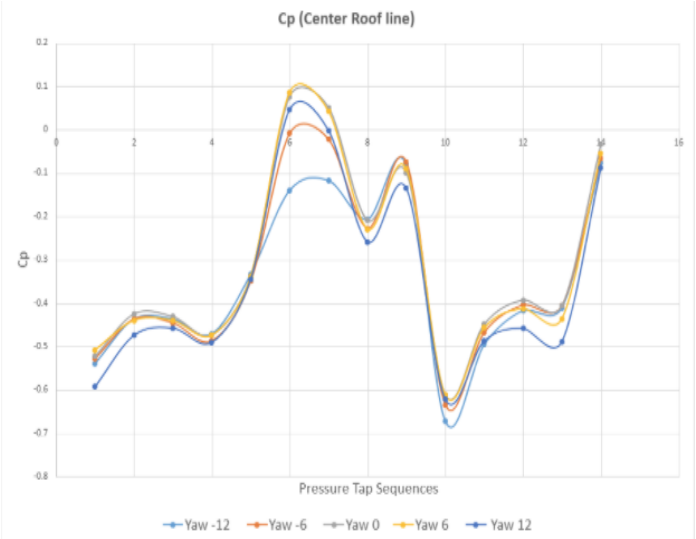


Figure 27: wind tunnel test results

Left roof line

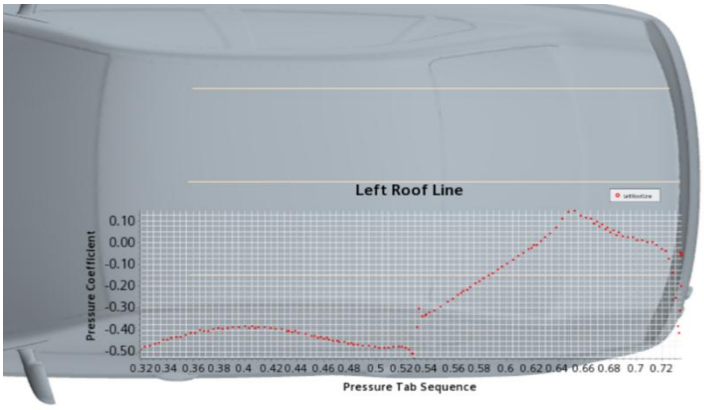
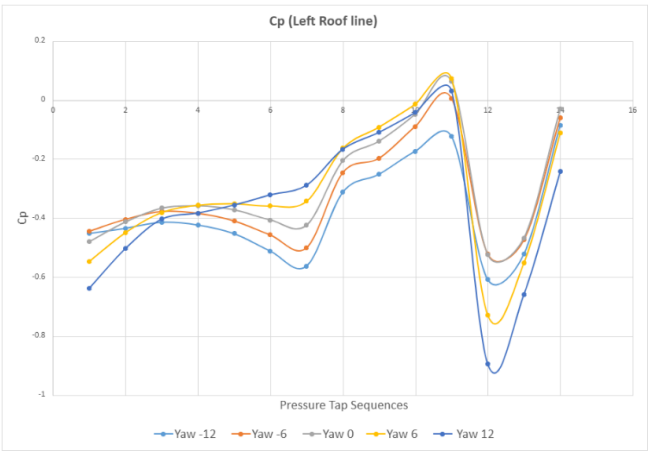


Figure 28: Star CCM result Figure



29: wind tunnel test results

Right roof line

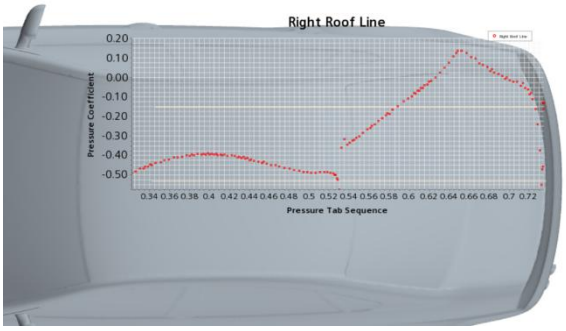


Figure 30: Star CCM result

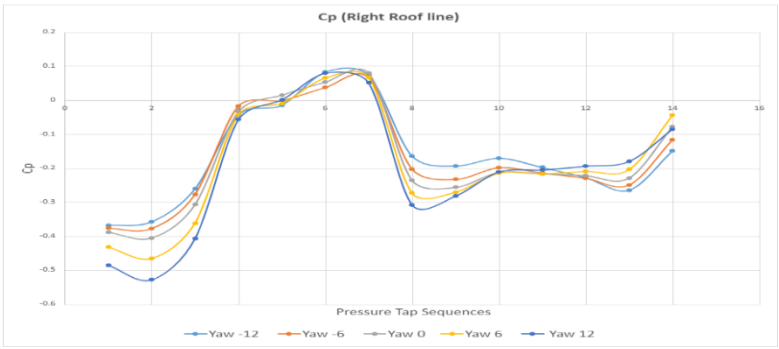


Figure 31: wind tunnel test results

When comparing the pressure coefficient distribution, it is evident that the CFD results generally capture the same trends. However, the CFD results sometimes predict slightly stronger suction specially near the rear end and roof regions. It suggests that separation intensity has been overestimated. The asymmetry observed under yaw conditions is also confirms that the model captures crossflow effects effectively. Overall, the CFD shows good qualitative agreement with the experimental data.

Comparing results between the two model configurations (with detailed and smooth underfloor)

Parameter	Smooth Underbody	Detailed Underbody
Cd	0.250477	0.304520
Cl	-0.08633	0.09302
CMy	-0.07353	-0.14192

Table 5: Smooth underbody and detailed underbody result comparison

Parameter	Smooth Underbody	Detailed Underbody	What it means
Cd	0.2505	0.3045	Drag is clearly higher for the detailed case. The extra geometry under the car disturbs the flow, causing more separation and a larger wake.
Cl	-0.0863	0.0930	Smooth underbody gives slight downforce, while the detailed one produces lift. This shows the smooth case has better underfloor flow control.
CMy	-0.0735	-0.1419	Both show nose-down behaviour, but the detailed case is much stronger. This suggests a shift in pressure distribution due to underbody disturbance.
Flow behaviour (overall)	Cleaner flow,	More disturbed	Underbody complexity increases separation and reduces aerodynamic efficiency.

	smaller wake	flow, larger wake	

Table 6: Analysis Summary (refer appendix A for detailed explanation)

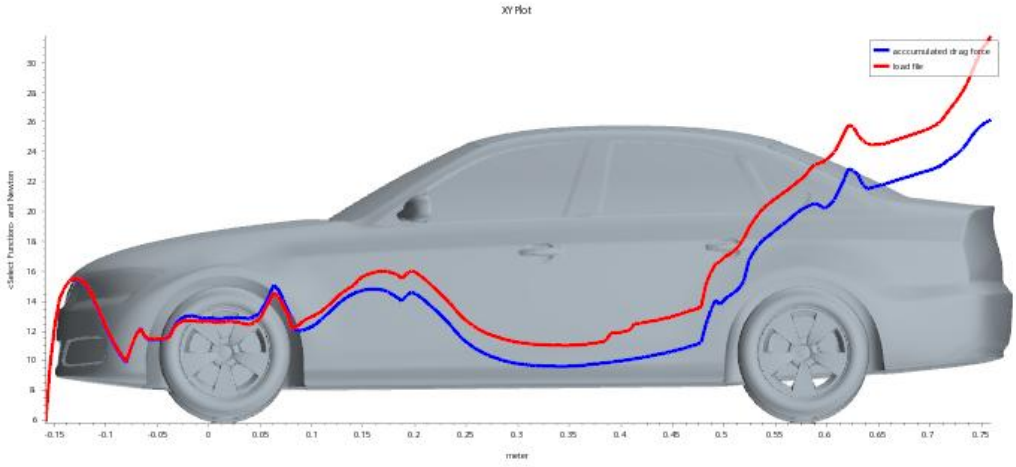


Figure 32: Detailed and Smooth model force plots

As per the results it is clear that detailed underbody (red) produces higher drag than the smooth underbody (blue) along most of the vehicle length. The largest difference can be seen in the rear.

In the front region, both configurations show similar drag levels. Reason for this is as per (Hucho, 2013) the flow is mainly controlled by stagnation pressure and initial boundary layer development. Those are less affected by underbody geometry (Hucho, 2013). However, moving along the underbody and mid-section, it can be seen that the detailed configuration generate higher drag. This is due to the complex geometry that disturbs the flow. It increases turbulence, and promote early boundary layer separation beneath the vehicle (Katz, 2016).

The largest difference can be seen in the rear region. The detailed underbody shows a sharp rise in drag when it comes to the rear end. As per (Heftet al. 2012) This behaviour is connected with stronger flow separation and a larger wake. This happens because of the disrupted underfloor flow interacting with the rear geometry. But the smooth underbody, allows more streamlined flow, reducing wake size and improving pressure recovery. And it leads to lower drag (Hucho, 2013).

Overall, the results show that underbody smoothness plays a major role in controlling flow separation, wake development, and aerodynamic efficiency. Since the smooth configurations reduces the drag by maintaining more stable and attached flow (Katz, 2016).

CFD Discussion

Flow Field Behaviour (Separation and Vortex Formation)

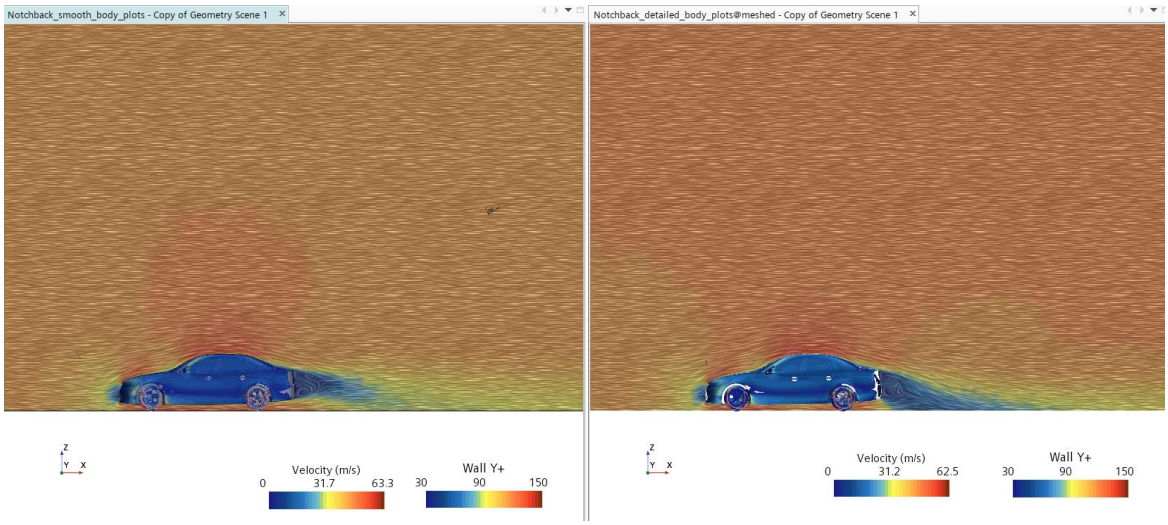


Figure 33: Vector flow distribution diagram (left - smooth underbody; right - detailed under body)

The flow visualisations indicate some differences between the smooth and detailed underbody configurations. In the smooth underbody design, the flow beneath the vehicle remains relatively uniform than the detailed underbody design. The streamlines are closely following the vehicle surface. And the wake region behind the vehicle is smaller and more controlled.

In the other hand the detailed underbody configuration shows more disturbed flow behaviour. The streamlines under the vehicle are less uniform, particularly around the wheel and underfloor regions. A strong velocity gradients and local disturbances are visible in those areas. These disruptions are caused by increased turbulence and earlier boundary layer separation. The wake region behind the vehicle is larger. A clear recirculation patterns can be seen and it suggests that a stronger vortex formation is there. This behaviour is normal for complex underbody geometries. Since it promotes flow separation and generates larger wake structures (Katz, 2016; Ahmed et al., 1984).

Overall, the detailed underbody produces stronger flow separation and more intense vortical structures, than the smooth underbody. Smooth underbody maintains better flow attachment and a more stable wake (Hucho, 2013).

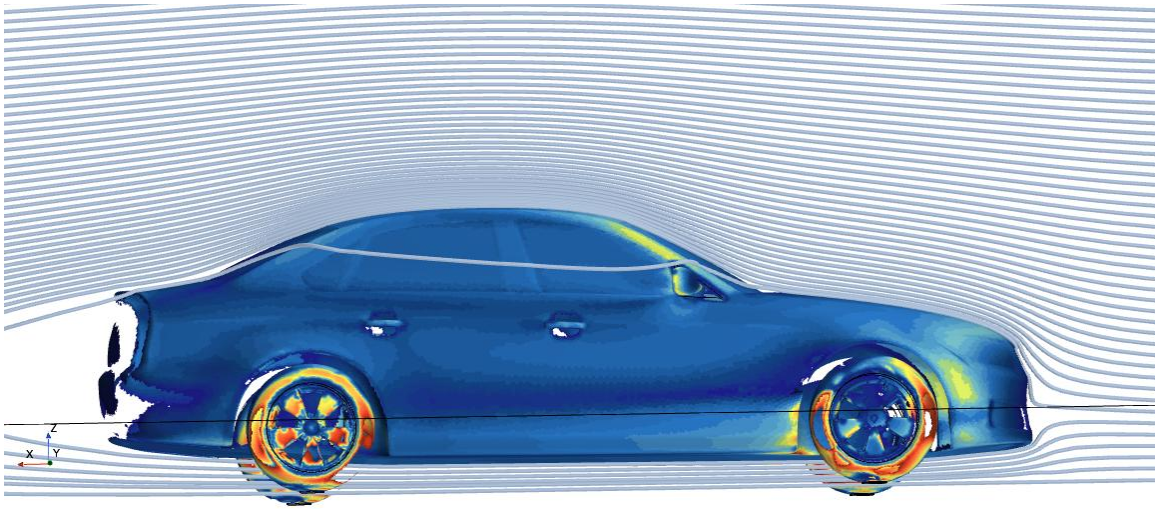


Figure 34: streamline stream for smooth underbody

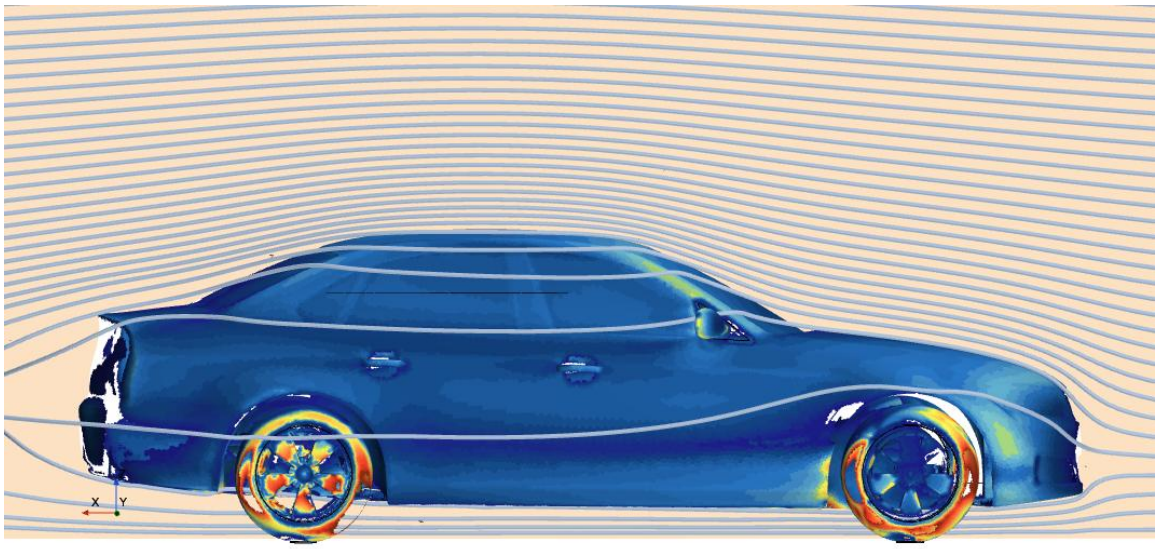


Figure 35: streamline stream for detailed underbody

Effect on Aerodynamic Performance and Optimal Setup

As per the explanations in above chapters it is clear that the smooth underbody configuration gives better aerodynamic performance than the detailed underbody. From a road vehicle perspective, an optimum aerodynamic setup should aim to minimise drag while maintaining stable lift and pitching characteristics.

It is clear that the smooth underbody is clearly closer to an optimal road setup. It provides an opportunity for more controlled underfloor flow while reducing flow separation, and limiting wake formation. It collectively improves aerodynamic efficiency and stability. In the other hand the detailed underbody has flow disturbances which increase separation and wake size finally resulting in reduced aerodynamic efficiency.

To further improve performance, modifications such as smoother underfloor panels, reduced exposure of underbody components, and improved rear-end flow management can be introduced (Hucho, 2013)

Conclusion

This study analyses the aerodynamic performance of the DrivAer model using wind tunnel testing and computational fluid dynamics (CFD) analysis. The findings proved that influence airflow dynamics around the vehicle is influenced by rear-end geometry and underbody design significantly.

From the different rear configurations, the Fastback gave the lowest drag and more stable flow, while the Estate showed the highest drag due to stronger separation and a larger wake. The Notchback behaved somewhere in between. When comparing CFD with experimental data, the overall trends matched well, particularly for lift and pitching moment, although drag was slightly overpredicted in the simulations.

The underbody comparison showed clear differences. The smooth underbody gave lower drag and even produced a small amount of downforce, which is generally beneficial. On the other hand, the detailed underbody disturbed the flow more, leading to higher drag, positive lift, and a stronger nose-down pitching effect. The flow visualisations also supported this, showing more separation and a larger wake for the detailed case.

Appendix A

Boundary conditions

The flow was modelled using Reynolds-Averaged Navier - Stokes (RANS) equations, as the high Reynolds number indicates fully turbulent flow. The main objective was to capture the mean aerodynamic behaviour efficiently without high computational cost. The SST k-ω turbulence model was selected due to its strong performance in predicting boundary layer behaviour, adverse pressure gradients, and flow separation, which are critical in vehicle aerodynamics, particularly around the rear and underbody regions. An all-high y^+ wall treatment was used to accommodate the chosen mesh resolution, providing flexibility for near-wall modelling under moderate prism layer conditions. It was assumed that the flow was steady, and incompressible as the velocity (45 m/s) is below the compressibility threshold.

For boundary conditions, slip walls selected to the wind tunnel boundaries to eliminate wall shear effects. This approach is appropriate since the domain size was sufficient have a low blockage ratio (<1%), minimising wall interference.

A Cartesian coordinate system was set to evaluate aerodynamic moments. The origin was set at (0, 0, 0.3 m), with standard longitudinal (X), lateral (Y), and vertical (Z) directions.

Reynolds Number Sweep Results

The Reynolds number is showing some clear aerodynamic differences which is caused by rear-end geometry. From the Cd vs Re results, the Fastback is the model which shows the lowest and most stable drag (≈ 0.23 - 0.26), and it is an indication of delayed separation and efficient pressure recovery as per (Hucho, 2013) and (Katz, 2016). The Estate model shows the highest drag, which is ≈ 0.30 at higher Reynolds numbers. This happens due to early separation and a large wake (Heft et al., 2012; Ahmed et al., 1984). The Notchback shows kind of an intermediate behaviour with small drag reduction at mid-range speeds but when it comes high speeds, it indicates a transition from partial flow attachment to wake-dominated flow.

For Cl, the Fastback model is maintaining low and stable values, while the Notchback model is showing higher lift. An it decreases slightly with Reynolds number. The Estate model depicts a reduction in lift while approaching near-zero or negative values at higher speeds which is consistent with strong rear separation (Hucho, 2013; Bearman, 1984).

The Pitch moment co efficient is negative for all models. It gives an indication that a nose-down tendency is there. The Fastback is mostly stable but the notchback becomes slightly more negative when the Reynolds number increases. Estate model shows the largest variation due to unstable wake effects (Heft et al., 2012; Katz, 2016).

Comparing results between the two model configurations

The results clearly show that underbody configuration impacts the aerodynamic performance. The drag coefficient (Cd) increased significantly from 0.2505 for the smooth underbody to 0.3045 when it comes to the detailed underbody. It indicates that the additional geometric complexity beneath the vehicle can create flow disturbances. In addition to that it enhances separation, and enlarges the wake region. It ultimately leads to higher aerodynamic resistance.

A notable difference can be seen in the lift coefficient (Cl) also. The smooth underbody produces a negative lift value (-0.0863), while detailed underbody results in a positive lift (0.0930). Negative value shows slight downforce effect. This indicates that the smooth configuration is promoting more favourable environment for underfloor flow acceleration and pressure reduction. But the detailed underbody won't let this happen easily, leading to a loss of downforce and increased lift.

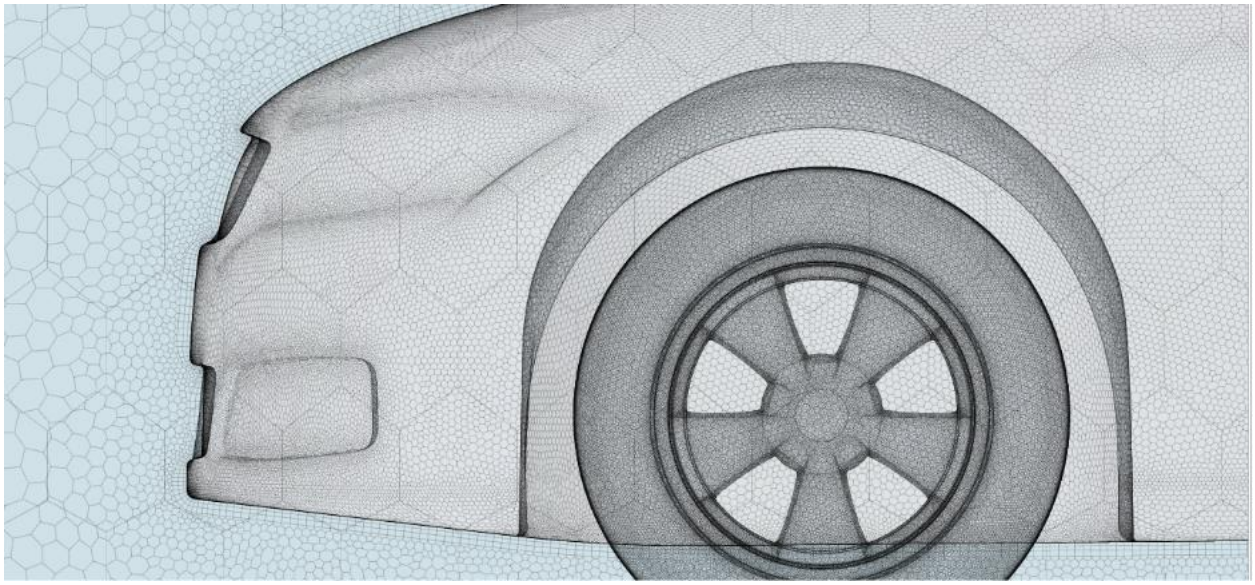
The pitching moment coefficient (CMy) also shows a change. The detailed underbody produces a higher negative value (-0.1419) compared to the smooth configuration (-0.0735). So that smooth body configuration indicates a significantly stronger nose-down pitching tendency. Altered pressure distribution along the underbody and rear region can be a reason for this. The increased flow separation shifts aerodynamic loading rearward

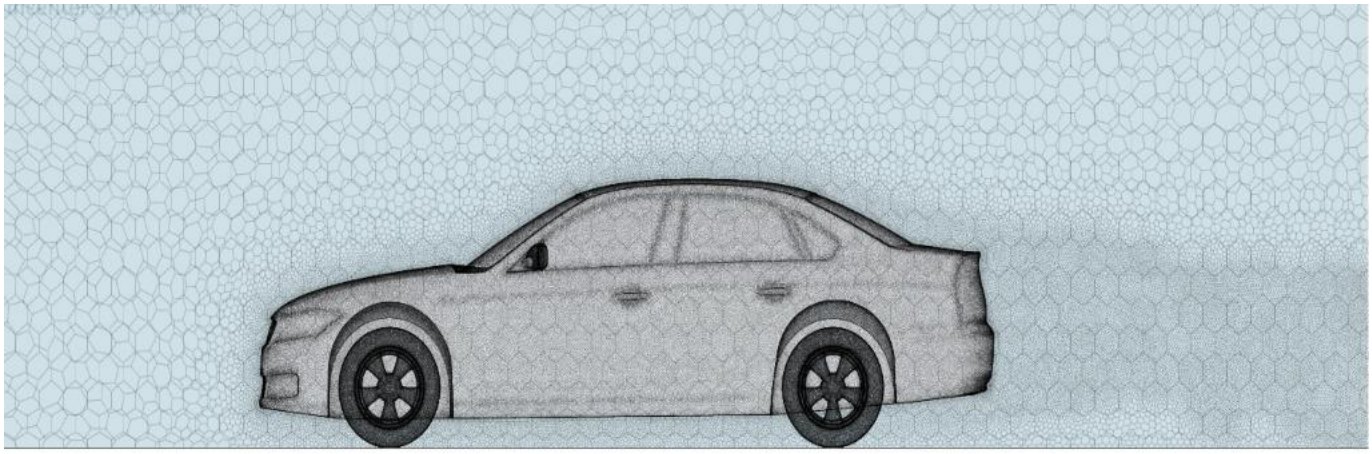
Comparing results with existing literatures

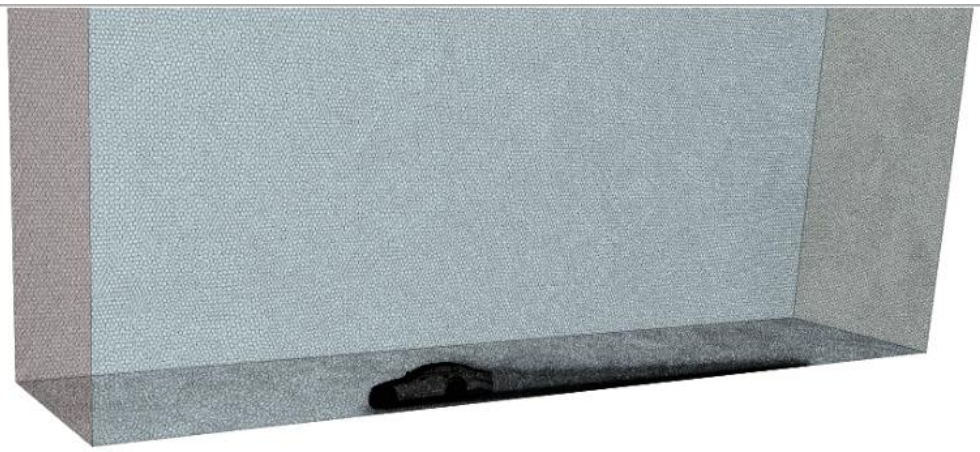
The CFD results for the Notchback configuration at 45 m/s and 0° yaw (Cd = 0.3045, Cl = 0.0930, CMy = -0.1419) show generally good agreement with values reported in the literature for similar automotive geometries. Studies on the DrivAer model and comparable notchback configurations indicate drag coefficients typically in the range of 0.26–0.29 (Heft et al., 2012; Hucho, 2013). The slightly higher drag predicted in the present study suggests that the CFD model may be overestimating wake size or flow separation, which is a common limitation of RANS-based simulations (Ekman et al., 2020).
The lift coefficient is consistent with literature, where notchback vehicles usually exhibit small positive lift due to pressure differences between the roof and underbody (Katz, 2016). Similarly, the negative pitching moment aligns with expected aerodynamic behaviour, indicating a stable nose-down tendency caused by rear-end separation effects.
Overall, the CFD results follow established aerodynamic trends, with minor discrepancies in drag prediction, while lift and pitching moment show strong agreement with published studies.

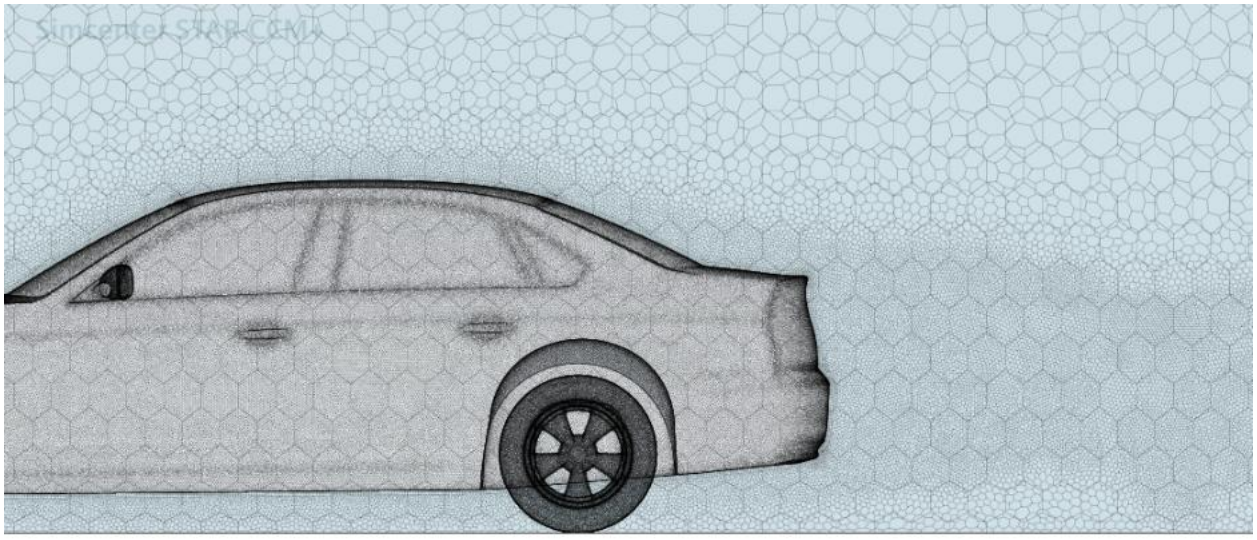
Appendix B

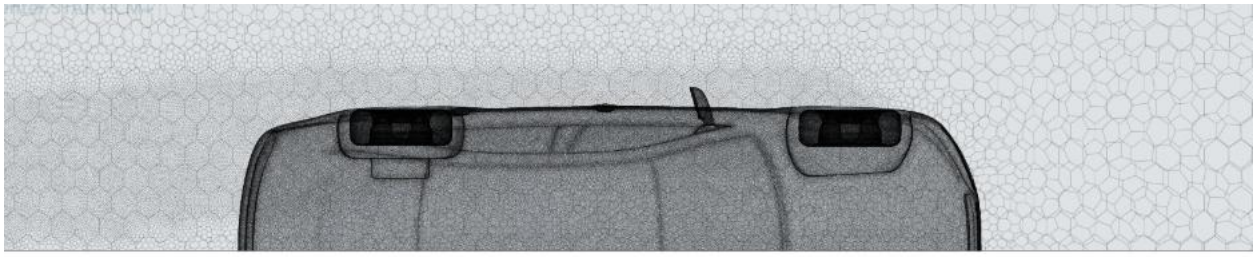
Restored views

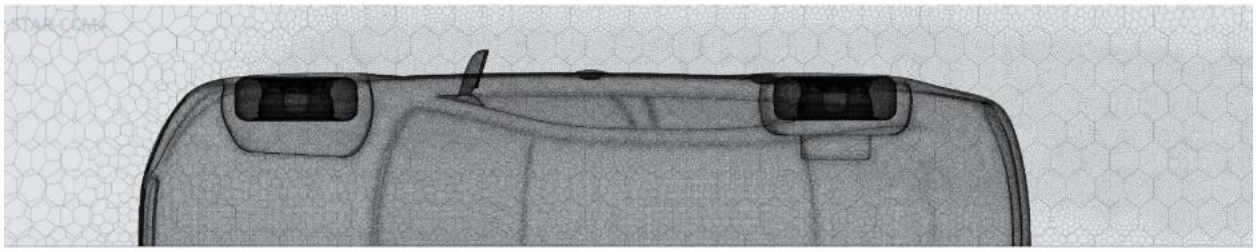


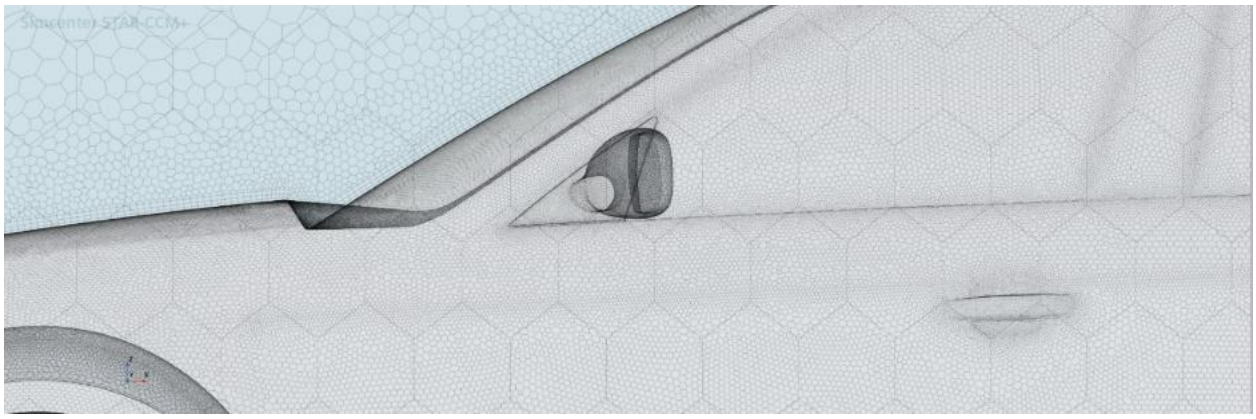


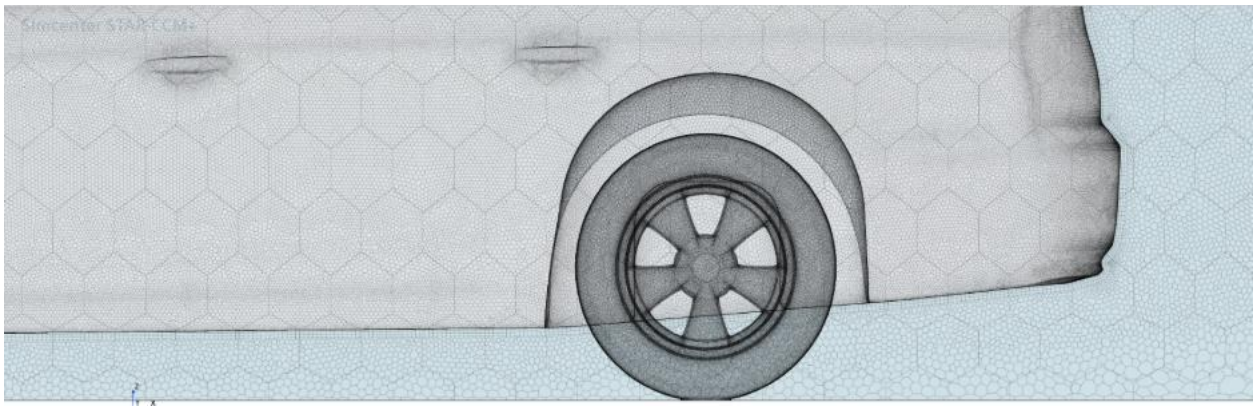



Simcenter STAR-CCM+









3. Vehicle Dynamics

3.1 Abstract

This study investigates the ride characteristics of a passenger vehicle using a 4DOF half-car model developed in MATLAB/Simulink. The model incorporates bounce and pitch motions of the sprung mass along with front and rear un-sprung masses, using representative vehicle parameters and mass distribution. A ride frequency of ***1.2 Hz (front)*** *and* ***1.44 Hz (rear)*** *is selected to achieve a balance between ride comfort and dynamic stability.*

The vehicle response to road excitation was analysed in both time and frequency domains. Frequency analysis using FFT identified a dominant peak at approximately ***1.2–1.3 Hz****, corresponding to the body bounce mode, and a secondary peak around* ***10–12 Hz****, associated with un-sprung mass (wheel-hop) dynamics. Ride comfort was evaluated using RMS acceleration of the sprung mass, which showed a reduction of approximately* ***20%*** *after optimisation of suspension damping.*

The results demonstrate that suspension damping plays a critical role in reducing vibration amplitudes and improving ride comfort, while stiffness primarily influences the natural frequency of the system.

Overall, the study highlights the importance of parameter tuning in achieving an optimal balance between comfort and vehicle dynamic performance

3.2 Vehicle Selection

The vehicle selected for this study is a Toyota Corolla 2.0L Hybrid (2019), which falls under the C - segment (medium passenger vehicle category) as specified in the assignment brief.

According to the classification provided, the medium segment includes vehicles with a mass range between 1100 kg and 1400 kg.

This vehicle was selected because it is a widely used passenger car, making it suitable for analysing ride comfort under typical driving conditions. Additionally, relevant vehicle parameters are easily available, which allows for a more realistic and reliable modelling approach.

3.3 Methodology

3.3.1 Ride Model

To perform the ride analysis on the selected vehicle, a half-car 4 degree-of-freedom (4DOF) model is used. This model allows the study of both bounce and pitch moments of the vehicle body, along with the dynamic behaviour of the front and rear unsprung masses.

The half-car model is chosen because it captures the interaction between the front and rear suspensions, which is essential when analysing ride comfort. When a vehicle passes over a bump, the front and rear wheels are excited at different times, resulting in pitch motion that significantly affects passenger comfort. Therefore, including pitch dynamics provides a more realistic representation of vehicle behaviour.

The model consists of four degrees of freedom:

- vertical displacement of the sprung mass z_s,
- front unsprung mass displacement z_{usf},
- rear unsprung mass displacement z_{usr},
- pitch θ.

This model was chosen because it captures the main dynamics that affect ride comfort, especially the vertical acceleration of the vehicle body, which is what passengers feel the most. The half-car model is commonly used in vehicle ride studies because it captures the interaction between front and rear suspension responses while remaining computationally efficient compared with a full-car model (Patole & Sawant, 2015; Tiwari & Bhavikatti, 2020).

3.3.2 Equations of Motion

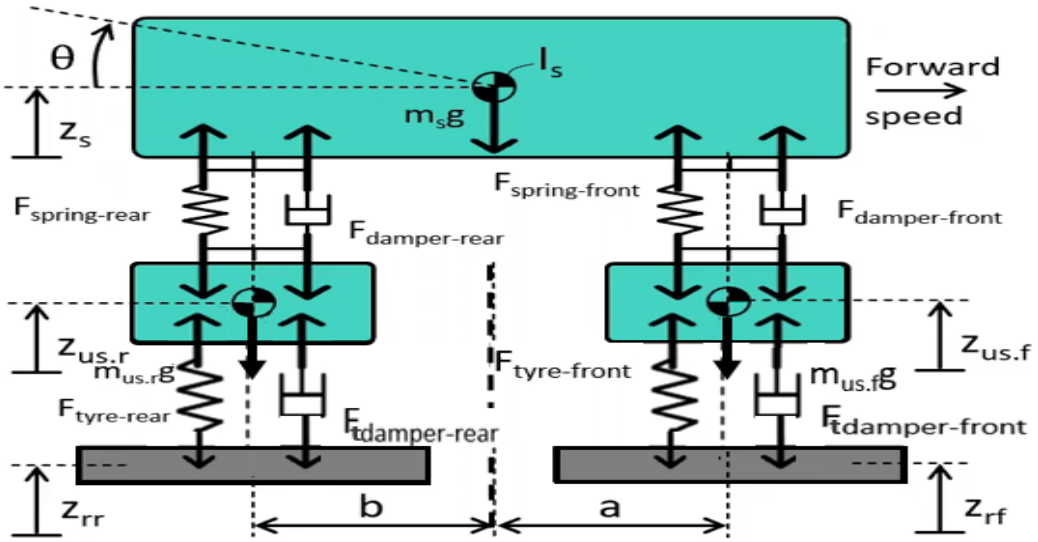


Figure 36: Free Body diagram of a 4DOF system

Using the above free body diagram from fig 33 the following equations of motion are derived:

The vertical motion of the vehicle body is represented by the following equation:

As in the quarter-car model, the un-sprung mass equilibrium can be determined and applied to the front and rear axles:

Front axle:

$$F_{tyre,f} - F_{spr,f} - F_{damp,f} - m_{us,f}g + F_{tdamp,f} = m_{us,f}\ddot{z_{us,f}}$$

Rear axle:

$$F_{tyre,r} - F_{spr,r} - F_{damp,r} - m_{us,r}g + F_{tdamp,r} = m_{us,r}\ddot{z_{us,r}}$$

The sprung mass vertical equilibrium is given by:

$$F_{spr,f} + F_{damp,f} + F_{spr,r} + F_{damp,r} - m_s g = m_s \ddot{z_s}$$

The sprung mass pitch equilibrium is obtained by taking moments about the centre of gravity:

$$-a\left(F_{spr,f} + F_{damp,f}\right) + b\left(F_{spr,r} + F_{damp,r}\right) = I_s\ddot{\theta}$$ refer fig 37

where:

- a and b are the distances from the centre of gravity to the front and rear axles, respectively
- θ is the pitch angle about the centre of mass (positive when the rear rises, negative when it lowers)

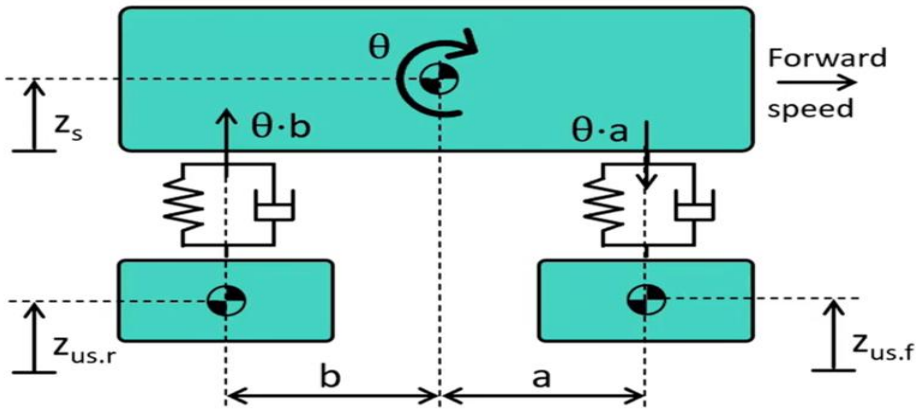


Figure 37: Free Body diagram of a 4DOF system for pitch movement

- As the vehicle pitch movement raises the rear and lowers the front under braking, the front axle suspension is compressed, while the rear is extended. The spring and damper forces are therefore as follows:

- Front Axle:

$$F_{spr,f} = F_{0,f} + k_{spr,f}\left(z_{us,f} - (z_s - \theta a)\right)$$

$$F_{damp,f} = c_{damp,f}\left(\dot{z_{us,f}} - (\dot{z_s} - \dot{\theta}a)\right)$$

- Rear Axle:

$$F_{spr,r} = F_{0,r} + k_{spr,r}\left(z_{us,r} - (z_s + \theta b)\right)$$

$$F_{damp,r} = c_{damp,r}\left(\dot{z_{us,r}} - (\dot{z_s} + \dot{\theta}b)\right)$$

where:

- **Tyre force (front/rear):** $F_{tyre} = k_{tyre}(z_{r,f/r} - z_{us,f/r})$
- **Spring force (front):** $F_{spr,f} = F_{0,f} + k_{spr,f}\left(z_{us,f} - (z_s - \theta a)\right), \quad F_{0,f} = \frac{m_s b}{L} g$
- **Spring force (rear):** $F_{spr,r} = F_{0,r} + k_{spr,r}\left(z_{us,r} - (z_s + \theta b)\right), \quad F_{0,r} = \frac{m_s a}{L} g$
- **Damper force (front):** $F_{damp,f} = c_{damp,f}\left(\dot{z_{us,f}} - (\dot{z_s} - \dot{\theta}a)\right)$
- **Damper force (rear):** $F_{damp,r} = c_{damp,r}\left(\dot{z_{us,r}} - (\dot{z_s} + \dot{\theta}b)\right)$
- **Tyre Damper force (front):** $F_{tdamp,f} = c_{tdamp,f}(\dot{z_{r,f}} - \dot{z_{us,f}})$
- **Tyre Damper force (rear):** $F_{rdamp,f} = c_{rdamp,f}(\dot{z_{r,r}} - \dot{z_{us,r}})$
- **Tyre equilibrium deflection (front):** $z_{0,tyre,f} = \frac{\left(m_s\frac{b}{L} + m_{us,f}\right)g}{k_{tyre}}$
- **Tyre equilibrium deflection (rear):** $z_{0,tyre,r} = \frac{\left(m_s\frac{a}{L} + m_{us,r}\right)g}{k_{tyre}}$
- F_0 is the "spring pre-load" - The force to overcome for the spring to act, this is used to adjust the ride height of a vehicle. This does not affect the stiffness of the system https://www.driftworks.com/articles/guides-featured-products/how-to-set-spring-preload-on-your-coilovers.html?utm_source

3.3.3 Parameter Definition

3.3.3.1 Vehicle Mass and CG calculation

The technical specifications of the selected vehicle indicate a wheelbase of **2.64 m** and a kerb mass range of **1385–1485 kg** for the 2.0-litre hybrid hatchback, making $M_{initial}$ = 1385 kg a suitable baseline value for this study (Toyota, n.d.).

- C.G location (without payload):

  Assuming a 60:40 front-rear weight distribution:

  $a_{initial}$ = 0.4*L = 0.4*2.64 = 1.056 m,

  $b_{initial}$ = L - $a_{initial}$ = 2.64 – 1.056= 1.584 m

- Axle Loads

  Total axle load: $W = Minitial \times g = 1385 \times 9.81 = 13586.85$ N

Front axle load: $W_f = \frac{W \cdot b}{L} = \frac{13586.85 \times 1.584}{2.64} = 8149.9$ N

Front axle load: $W_r = \frac{W \cdot a}{L} = \frac{13586.85 \times 1.056}{2.64} = 5436.9$ N

- Unsprung Mass Distribution

  Assuming 10% of the total mass in unsprung mass:

  $m_u = 0.1 \times 1385 = 138.5$ kg

  Front unsprung mass: $m_{usf} = 138.5 * 0.5 = 69.25$ kg

  Rear unsprung mass: $m_{usr} = 138.5 * 0.5 = 69.25$ kg

- Sprung Mass Distribution

  Total sprung mass: $m_{s=} M_{initial} - m_{usf} - m_{usr} = 1246.5$ kg

  Front sprung mass: $m_{sf} = 1246.5 \times \frac{1.584}{2.64} = 747.9$ kg

  Rear sprung mass $m_{sr} = 1246.5 \times \frac{1.056}{2.64} = 498.6$ kg

3.3.3.2 Suspension Parameters

- Ride Frequency
  - The ride frequency ratio is described by the relationship between the front and rear suspension ride frequencies.
  - For passenger vehicles, there is a preference to have a bounce motion instead of a pitching motion, and so the rear suspension ride frequency is typically higher than the front axle. Typically, $f_{rear} \approx 1.2 * f_{front}$
  - Rear suspension ride frequency should be higher than the front for the rear axle to 'catch up' after a bump.

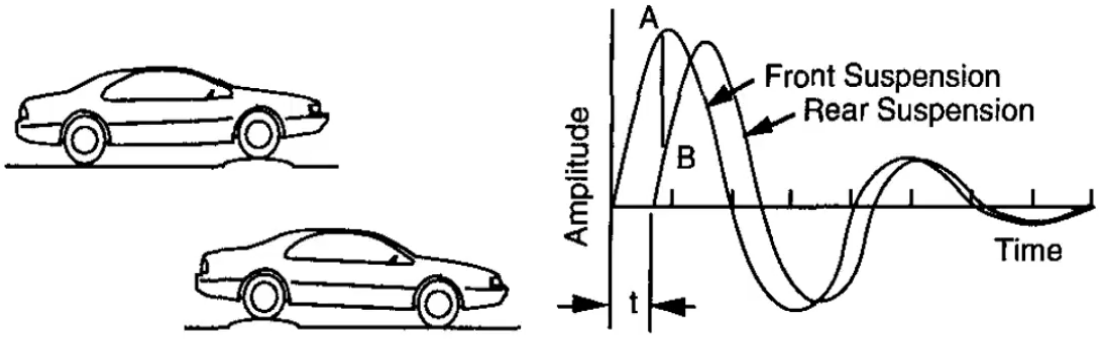


Figure 38: Road Profile Input

Source: T. Gillespie – Fundamentals of Vehicle Dynamics

- Spring Stiffness

Using the axle load and chosen ride frequencies, the front and rear suspension stiffness values are calculated from the standard natural frequency formula:

$$k = \omega_n^2 m$$

where $\omega_n = 2\pi f_n$.

Front suspension stiffness: $k_{spr,f} = (2\pi \times 1.24)^2 \times 747.9 = 45600\ \mathrm{N/m}$

Rear suspension stiffness: $k_{spr,r} = (2\pi \times 1.488)^2 \times 498.6 = 43600\ \mathrm{N/m}$

- Damping Coefficients

Similarly, the critical damping coefficients were calculated from:

$$c_c = 2\zeta\sqrt{km}$$

This method is appropriate for a passive ride model because it produces representative suspension parameters from mass and frequency targets, which is acceptable where manufacturer suspension data are not directly available in the public domain (assignment brief; Tiwari & Bhavikatti, 2020).

Assuming damping ratio $\zeta = 0.2$

Front critical damping: $c_{cf} = 2\sqrt{45600 \times 747.9} = 11670\ \mathrm{Ns/m}$

Front suspension damping: $c_{damp,f} = 0.2 \times 11670 = 2334\ \mathrm{Ns/m}$

Rear critical damping: $c_{cr} = 2\sqrt{43600 \times 498.6} = 9320\ \mathrm{Ns/m}$

Rear damping coefficient: $c_{damp,r} = 0.2 \times 9320 = 1864\ \mathrm{Ns/m}$

3.3.3.3 Tyre Parameters

Tyre stiffness is assumed as:

$k_{tyre} = 290000\ N/m$

Considering tyre modelling as a linear spring, static deflection is calculated using equilibrium relation between load and stiffness

$$z_o \frac{W}{k_{tyre}}$$

This approach assumes that the tyre behaves as a linear elastic element under vertical loading (Gillespie, 1992)

Front tyre deflection: $z_{0f} = \frac{(747.9+69.25)\times 9.81}{290000} = 0.028\ \mathrm{m}$

Rear tyre deflection: $z_{0r} = \frac{(498.6+69.25)\times 9.81}{290000} = 0.019\ \mathrm{m}$

Parameter	Symbol	Value	Unit
Wheelbase	L	2.64	m
Vehicle mass	M	1385	kg
C.G location from front axel	$a_{initial}$	1.056	m
C.G location from rear axel	$b_{initial}$	1.584	m
Pitch Inertia	$I_{initial}$	1728.758	$Kg.m^2$

Table 7: Vehicle Parameters (Without Payload)

Parameter	Symbol	Value	Unit
Sprung mass	m_s	1246.5	kg
Front sprung mass	m_{sf}	747.9	kg
Rear sprung mass	m_{sr}	498.6	kg
Front unsprung mass	m_{usf}	69.25	kg
Rear unsprung mass	m_{usr}	69.25	kg

Table 8: Mass Distribution (Without Payload)

Parameter	Symbol	Value	Unit
Front ride frequency	f_{front}	1.24	Hz
Rear ride frequency	f_{rear}	1.488	Hz
Front stiffness	$k_{spr,f}$	45600	N/m
Rear stiffness	$k_{spr,r}$	43600	N/m
Front damping ratio	ζ_f	0.2	-
Rear damping ratio	ζ_r	0.2	-
Front damping coefficient	$c_{damp,f}$	2334	Ns/m
Rear damping coefficient	$c_{damp,r}$	1864	Ns/m

Table 9: Suspension Parameters (Without Payload)

Parameter	Symbol	Value	Unit
Tyre stiffness	k_{tyre}	290000	N/m
Tyre damping coefficient	$c_{tdamp,f}$	121	Ns/m
Front tyre deflection	z_{0f}	0.028	m
Rear tyre deflection	z_{0r}	0.019	m

Table 10: Tyre Parameters (Without Payload)

3.3.4 Payload and Weight Distribution

3.3.4.1 CG Change

Before payload was added, the model assumed a 60:40 front-to-rear weight distribution. The payload shifted the centre of gravity according to the passenger position, and luggage position. The updated location is calculated in MATLAB using the following formula:

- C.G location

From Table1 $a_{initial} = 1.056$

$$a = \frac{M_{initial}(a_{initial}) + mp(xp) + ml(xl)}{M} = 1.0186\,m$$

$$b = L - a = 2.64 - 1.0186 = 1.6214$$

- Axle Loads

$$W_f = \frac{Wb}{L} = \frac{14460 \times 1.6214}{2.64} = 8874.7\ N$$

$$W_r = \frac{Wa}{L} = \frac{14460 \times 1.0186}{2.64} = 5579.15\ N$$

- Unsprung Mass Distribution

  Assuming 10% of the total mass in unsprung mass:

  $$m_u = 0.1 \times 1474 = 147.4 \text{ kg}$$

  Front unsprung mass: $m_{usf} = 147.4 * 0.5 = 73.7$ kg

  Rear unsprung mass: $m_{usr} = 147.4 * 0.5 = 73.7$ kg

- Sprung Mass Distribution

  Total sprung mass: $m_s = M - m_{usf} - m_{usr}$=1326.6 kg

  Front sprung mass: $m_{sf} = 1326.6 \times \frac{1.6214}{2.64} = 814.75\ kg$

  Rear sprung mass: $m_{sr} = 1326.6 \times \frac{1.0186}{2.64} = 511.85$ kg

This change in CG location is significant because front and rear axle loads determine the suspension stiffness values calculated for each end of the car. Since the front axle carries more load, the front suspension stiffness is greater than the rear stiffness in the baseline setup. This is physically reasonable and helps the front suspension support the heavier end of the vehicle. It also affects the pitch response, because front and rear suspensions no longer respond symmetrically to the same road disturbance.

3.3.4.2 Inertia Change

The added payload, i.e. 89kg, increases the total vehicle mass from 1385 kg to 1474 kg. The final two digits of the group leader's SID are 89, which were distributed as 79 kg passenger mass and 10 kg luggage mass. The passenger is positioned at 0.2 m from the front axle, and the luggage at 2.3 m from the front axle. These locations are reasonable because they approximately represent the driver at the front seat and luggage loaded in the boot. This increase also changes the vehicle pitch inertia. Inertia after adding the payload is calculated by the following formula:

$I = I_{initial} + mp(xp - a)^2 + ml(xl - a)^2$ where;

Initial Inertia before adding payload	$I_{initial} = 1728.758\ kg \cdot m^2$
Passenger Mass	$mp = 79$ kg,
Passenger Position	$xp = 0.2$ m,
Position of CG from the front axle	$a = 1.0186$ m,
Luggage Mass	$ml = 10$,
Luggage Position	$xl = 2.3$ m,

$$I = 1728.758 + 79\,(0.2 - 1.0186)^2\,10\,(2.3 - 1.0186)^2 = 1798.116$$

The pitch inertia increased from 1728.758 kg·m² to 1798.116 kg·m², reflecting the effect of the loaded vehicle to angular acceleration about the pitch axis.

This should be taken into consideration because pitch inertia directly influences how strongly the body rotates when the front and rear wheels encounter the bump at different times. A larger pitch inertia generally reduces the rate of body rotation, but it also changes the transient response. This relationship between bounce-pitch coupling and vehicle parameters is well recognised in half-car ride analysis (Zhang et al., 2016)

Parameter	Symbol	Value	Unit
Wheelbase	L	2.64	m
Vehicle mass	M	1474	kg
C.G location from front axel	$a_{initial}$	1.0186	m
C.G location from rear axel	$b_{initial}$	1.6214	m
Pitch Inertia	I	1798.116	$Kg.m^2$

Table 11: Vehicle Parameters (With Payload)

Parameter	Symbol	Value	Unit
Sprung mass	m_s	1326.6	Kg
Front sprung mass	m_{sf}	814.75	Kg
Rear sprung mass	m_{sr}	511.85	kg
Front unsprung mass	m_{usf}	73.7	kg
Rear unsprung mass	m_{usr}	73.7	kg

Table 12: Mass Distribution (With Payload)

Parameter	Symbol	Value	Unit
Front ride frequency	f_{front}	1.24	Hz
Rear ride frequency	f_{rear}	1.488	Hz
Front stiffness	$k_{spr,f}$	49600	N/m
Rear stiffness	$k_{spr,r}$	44700	N/m
Front damping ratio	ζ_f	0.2	-
Rear damping ratio	ζ_r	0.2	-
Front damping coefficient	$c_{damp,f}$	2546	Ns/m
Rear damping coefficient	$c_{damp,r}$	1912	Ns/m

Table 13: Suspension Parameters (With Payload)

Parameter	Symbol	Baseline Value	With Payload	Change	Unit
Total mass	M	1385	1474	+89	kg
CG (front to CG)	A	1.056	1.0186	- 0.4586	m
CG (CG to rear)	B	1.584	1.6214	+0.0284	m
Front axle load	Wf	8149.9	8874.7	+724.8	N
Rear axle load	Wr	5436.9	5579.15	+142.25	N
Sprung mass	m_s	1246.5	1326.6	+80.1	kg
Front sprung mass	m_{sf}	747.9	814.75	+66.85	kg
Rear sprung mass	m_{sr}	498.6	511.85	+13.25	kg
Front stiffness	$k_{spr,f}$	45600	49600	+4000	N/m
Rear stiffness	$k_{spr,r}$	43600	44700	+1100	N/m
Front damping	$c_{damp,f}$	2334	2546	+212	Ns/m
Rear damping	$c_{damp,r}$	1864	1912	+48	Ns/m
Pitch inertia	I	1728.8	1780.5	+51.7	kg·m²

Table 14: Comparison Table (Baseline vs Payload)

3.4 Model Implementation

3.4.1 Road Input

- Half Sine Bump

  The road input used in the Simulink model was a half sine bump applied first to the front wheel and then to the rear wheel after a wheelbase delay. Considering the delay is necessary in a half-car ride model because the front and rear wheel excitation is what generates pitch motion (Patole & Sawant, 2015).

  The input values were:

  - Vehicle speed, $v = 4$ m/s
  - Bump height, $h = 0.03$ m
  - Bump length, $l = 2$ m
  - Wheelbase, $L = 2.64$ m
  - Front wheel bump start time, $t_0 = 0.5$ s

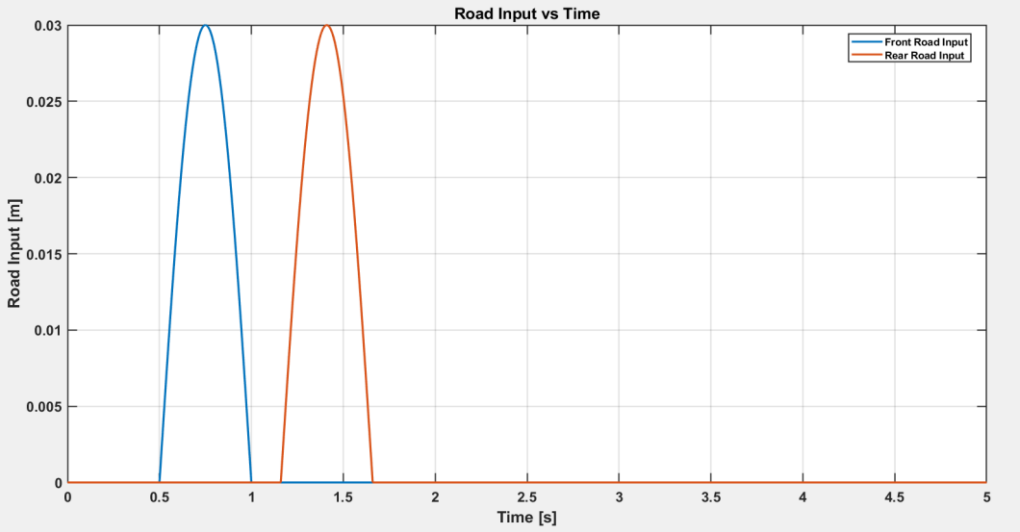


Figure 39: Road Input [m] vs Time [s]

The bump duration was calculated as:

$$T = \frac{l}{v} = \frac{2}{4} = 0.5 \text{ s}$$

The rear wheel delay was calculated from the wheelbase and vehicle speed:

$$\Delta t = \frac{L}{v} = \frac{2.64}{4} = 0.66 \text{ s}$$

so the rear wheel encountered the bump at:

$$t_1 = t_0 + \frac{L}{v} = 1.16 \text{ s}$$

This type of excitation is important because it allows the model to reproduce both heave and pitch responses of the vehicle body.

- Frequency Sweep input

  A frequency sweep (chirp) signal is used as the road excitation input for dynamic response of the vehicle model over wide range of frequencies. Input signal varies from low frequency to a high frequency over a specified duration. Excitation freq increase from 0.1 Hz to 50 Hz over a total time simulation of T = 100 sec.

➢ Input signal Definition

Chirp signal is applied with a constant velocity amplitude, and the displacement input is obtained by integrating the velocity signal



Figure 40: Chirp Signal Input Simulink model

This approach ensures that:

- At low frequencies the resulting displacement amplitude is large
- At high frequencies the displacement amplitude becomes small as shown in fig

Relation between excitation frequency and road wavelength is given by $f = \frac{V}{\lambda}$

Wavelength	Frequency	Physical Meaning
Large λ	Low f	Big bumps / undulations
Small λ	High f	Rough surface

Table 15: Relationship Between Road Wavelength and Excitation Frequency

Low frequency excitation corresponds to long-wavelength road disturbance such as speed bumps and large undulations, whereas high frequency excitations correspond to short wavelength surface irregularities.

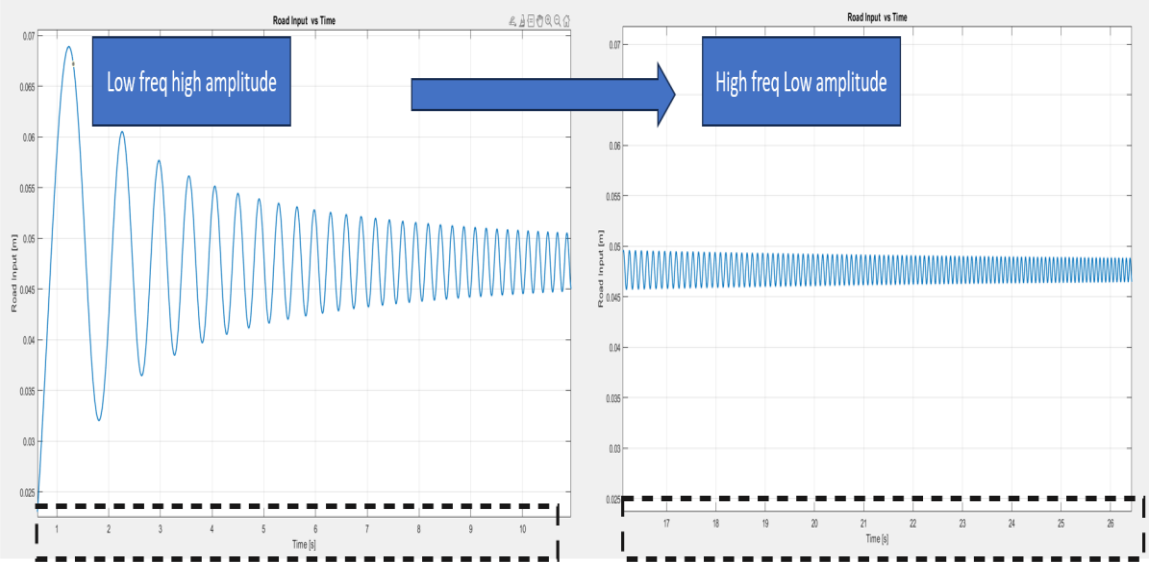


Figure 41: Frequency Sweep input3.4 Results and Discussion

3.4.2 Baseline Results

- **Half Sine Bump Analysis**

  The baseline simulation represents the vehicle travelling over a 30 mm bump at 4 m/s using the suspension parameters defined earlier. In a half-car model, the most important outputs for ride assessment are the sprung-mass vertical displacement, sprung-mass vertical acceleration, pitch angle, and the motions of the front and rear unsprung masses. These outputs indicate how strongly the vehicle body responds to the road input and how effectively the suspension isolates the cabin (Tiwari & Bhavikatti, 2020).

  When the front wheel first encounters the bump, the front unsprung mass rises rapidly, as the tyre follows the road input immediately. The front suspension then transmits disturbance to the sprung

mass, producing an initial body motion. When the rear wheel reaches the same bump after the wheelbase delay, a second body motion occurs. This delayed rear excitation creates the bounce pitch coupling seen in half-car models (Patole & Sawant, 2015; Zhang et al., 2016).

The most important output is the sprung mass acceleration, since this is the quantity most directly felt by passengers. Figure 43 shows the vertical acceleration response of the sprung and unsprung masses as the vehicle passes over the bump. The front unsprung mass experiences the first peak (≈7.8 m/s²) at about 0.52 s, followed by the rear unsprung mass (≈8.4 m/s²) at about 1.18 s due to the wheelbase delay. The sprung mass acceleration remains much lower (≈1.37 m/s²), indicating effective isolation of the vehicle body from road disturbance. The oscillations smoothen gradually, showing that the selected parameters provide stable control

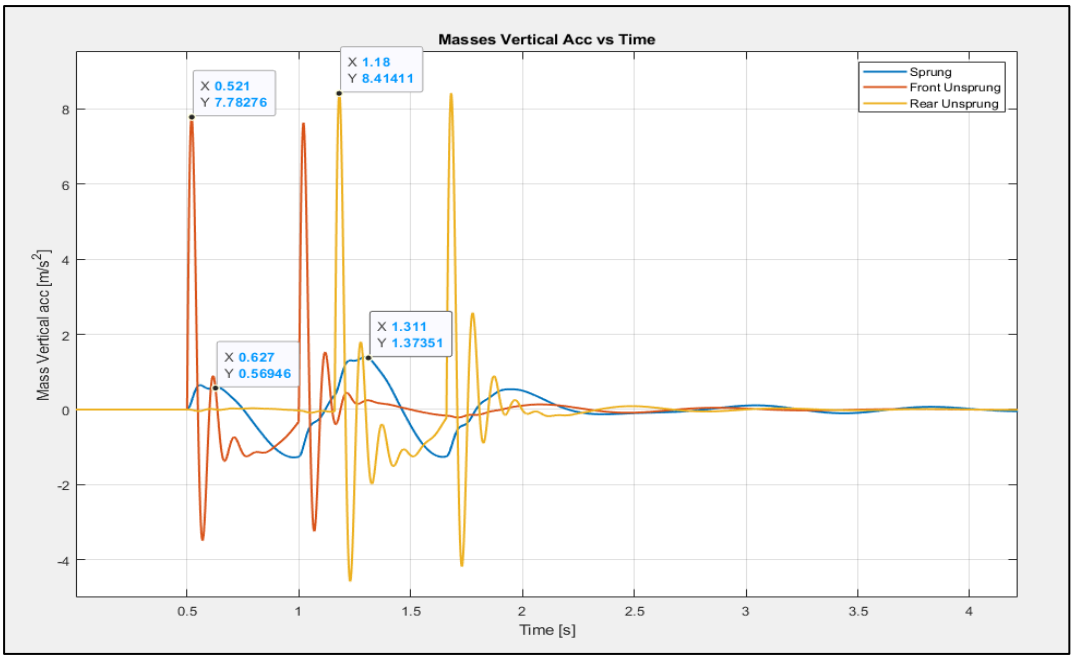


Figure 42: Vertical acceleration $[m/s^2]$ of sprung and unsprung masses vs Time[s]

Figure 43 shows the vertical displacement of the sprung and unsprung masses together with the road inputs. The front wheel follows the bump profile first (≈0.031 m at 0.76 s), followed by the rear wheel at about 1.44 s. The sprung mass displacement is smaller (≈0.026 m), confirming that the suspension reduces disturbance before it reaches the vehicle body. The response then smooths out.

To further evaluate ride comfort, the root mean square (RMS) acceleration of the sprung and unsprung masses is calculated. The RMS value for the sprung mass acceleration is:

$$0.0969 \text{ m/s}^2$$

This value is well below the threshold of 0.315 m/s², which corresponds to a comfortable ride according to ISO 2631. This indicates that the baseline suspension setup provides a good level of ride comfort, as the vibrations transmitted to the vehicle body are minimal (ISO 2631-1, 1997). The RMS values for the unsprung masses were:

$$\text{Front: } 0.1952 \text{ m/s}^2$$

$$\text{Rear: } 0.2234 \text{ m/s}^2$$

These values are higher than the sprung mass acceleration, which is expected since the wheels are directly exposed to the road input. The slightly higher RMS value at the rear is due to the delayed excitation caused by the wheelbase, resulting in a second disturbance input to the system. Overall, the baseline setup shows a comfortable and stable response, with controlled vibrations and effective isolation of the vehicle body. However, it can still be improved, particularly in reducing peak accelerations and improving the rate at which oscillations decay

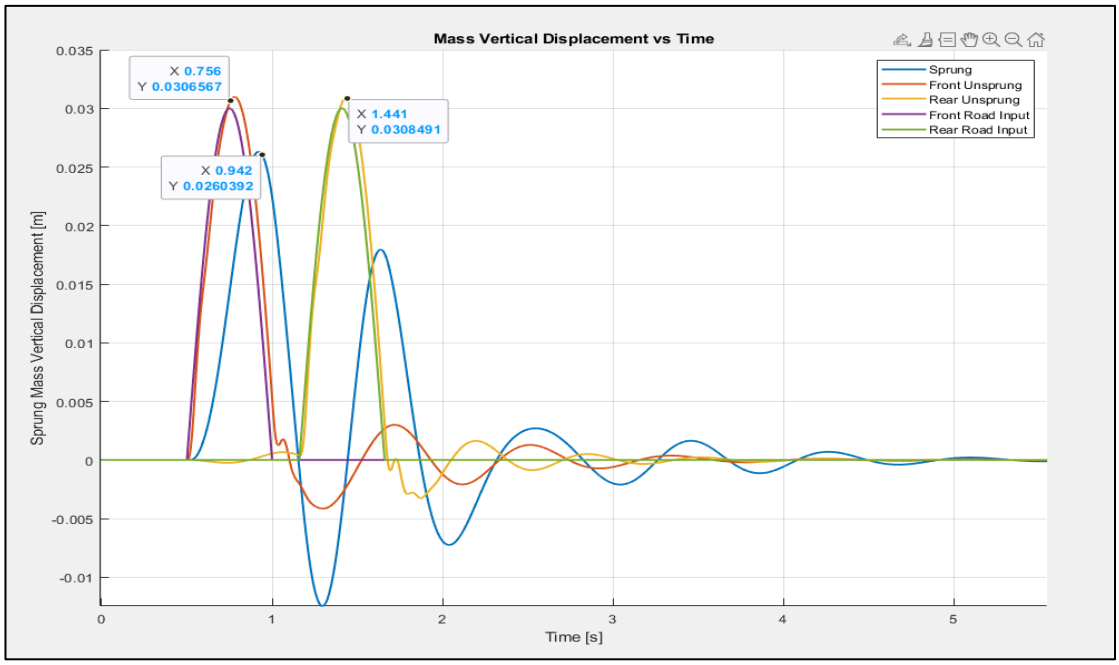


Figure 43: Vertical Displacement[m] of sprung and unsprung masses vs Time[s]

- **Frequency Analysis**

Frequency response of model is analysed using chirp input as discussed in the above section of road input.

From sprung mass acceleration plot shows a clear peak approximately at 1.21 Hz, which corresponds to the body bounce mode. Primary mode affecting the ride comfort, as it represents the vertical motion of the vehicle body.

In figure 42 Second peak is observed in unsprung mass response which is at 10.643 Hz for front and 10.71 Hz for the rear, value corresponds to wheel hop mode. This is governed by tyre stiffness and unsprung mass and represents the vibration of the wheel assembly.

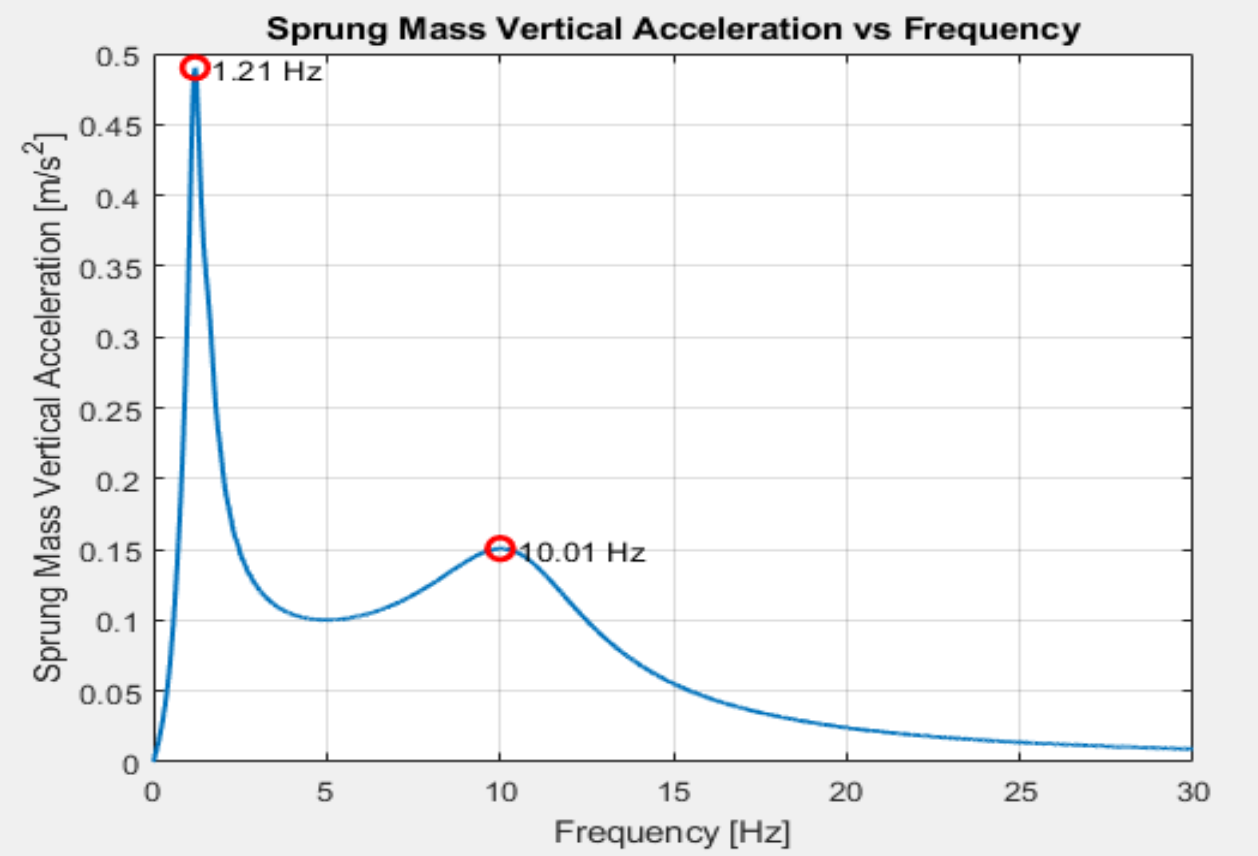


Figure 44: Sprung mass Vertical Acceleration vs Frequency

Results indicate a separation between body and wheel dynamics. Low frequency region associated with ride comfort while high frequency region associated with road holding.

Analytically calculated wheel hop frequencies are 10.8 Hz and 10.72 Hz for front and rear respectively which closely match the simulation result, which validates the Simulink model.

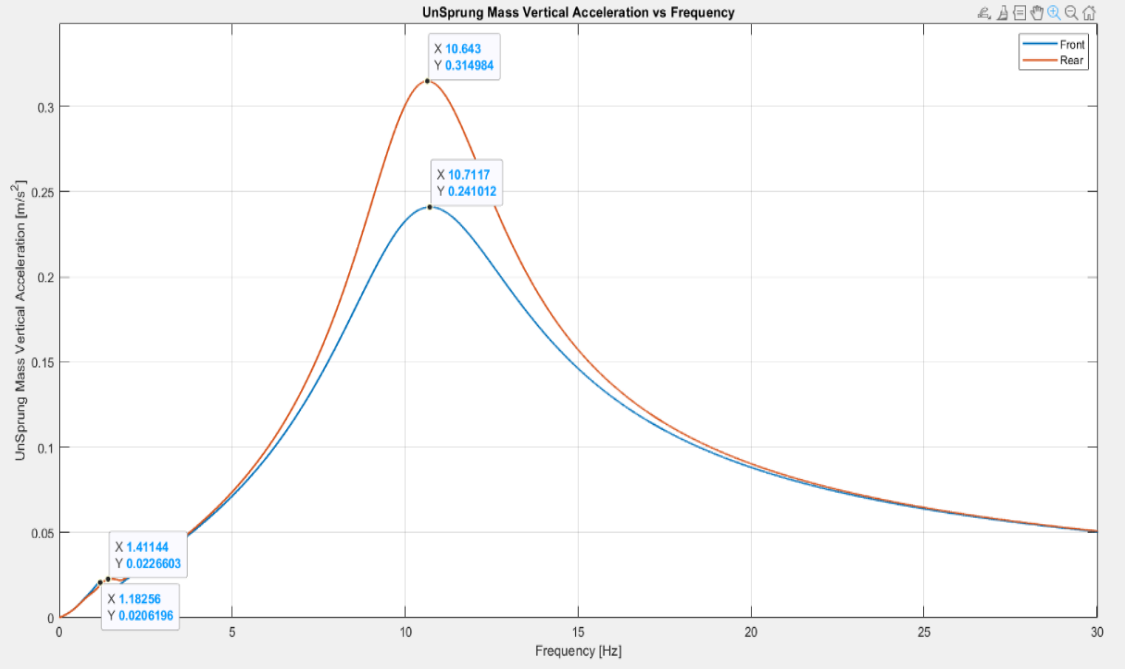


Figure 45: Unsprung Mass Vertical Acceleration vs Frequency

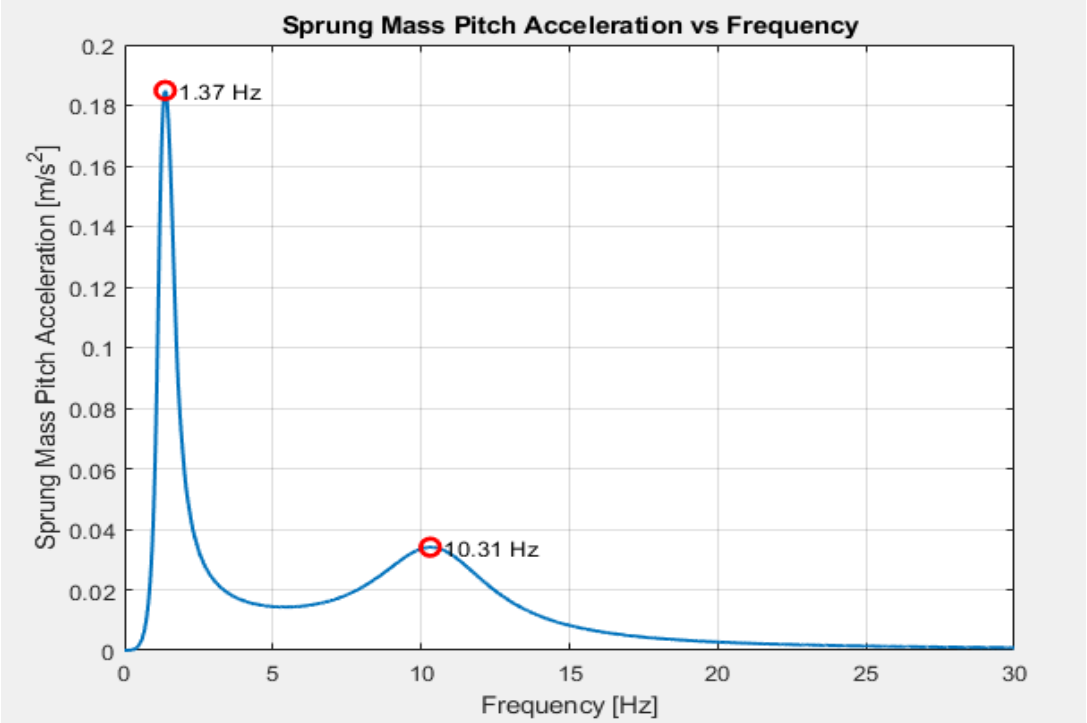


Figure 46: Unsprung Mass Vertical Acceleration vs Frequency (ff=1.24, $\zeta f = 0.2$, $\zeta r = 0.2$*)*

From figure 46, as the frequency increases, amplitude rise sharply with the highest amplitude at 1.37 Hz which represent the natural frequency of the vehicle. Resonance is seen due to interaction between the front and rear suspension system.

3.4.3 Parameter Study

Study is done to investigate the combined effect of suspension stiffness and damping ratio on ride comfort. Suspension stiffness is varied by changing the ride frequency in two stages first it was varied from 0.9 Hz to 1.75 Hz, for second stage the frequency range is narrowed down between 1.14 Hz to 1.24 Hz while damping ratio is varied from 0.2 to 0.5.

The performance evaluation of the system is done using RMS value of sprung mass acceleration, which is the key indicator of ride comfort.

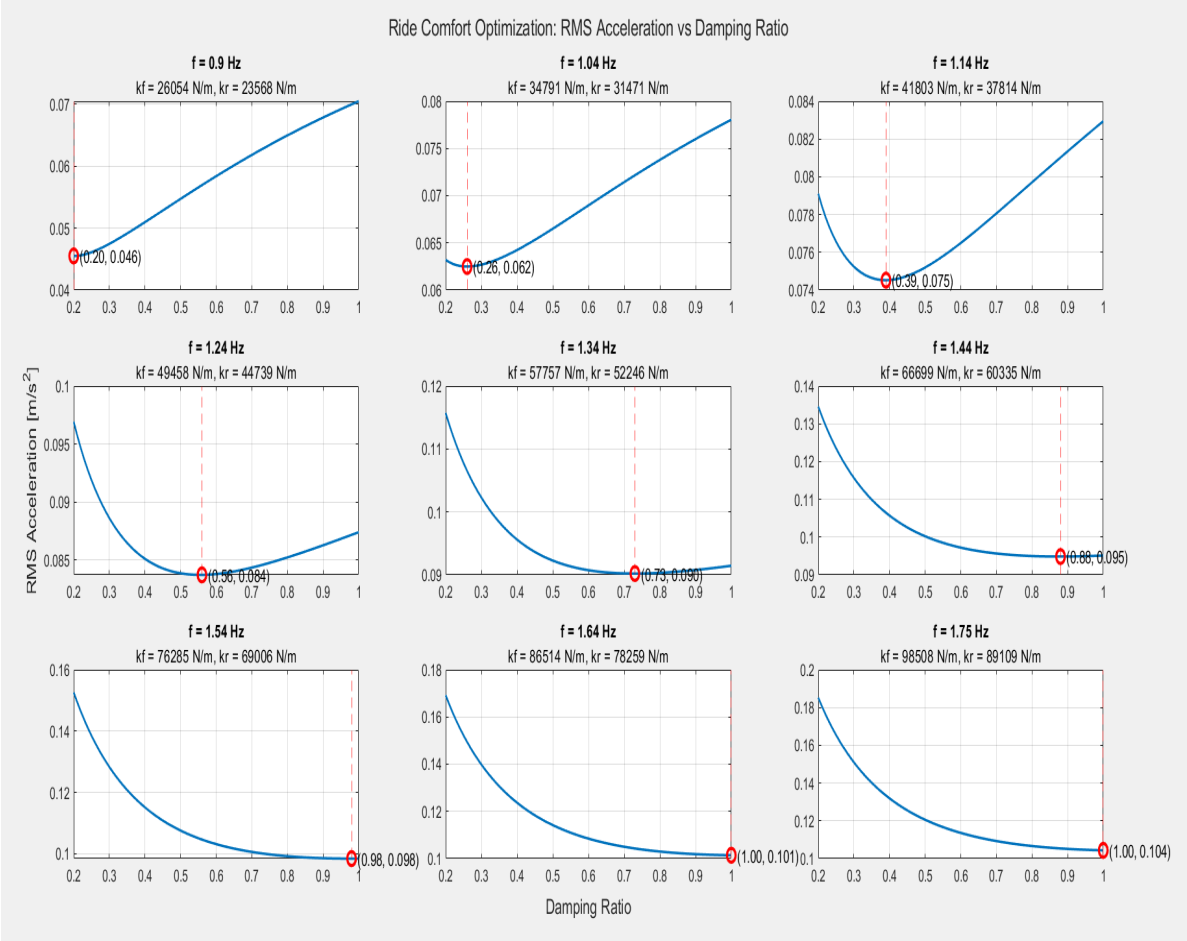


Figure 47: Wide Frequency Range (0.9 Hz to 1.75 Hz)

- Wide Frequency Range (0.9 Hz to 1.75 Hz):
    - At low ride frequencies (0.9 Hz to 1.1 Hz):

      Suspension is softer, resulting in lower rms acceleration and improved comfort.
    - At high ride frequencies (>1.5 Hz):

      Suspension becomes significantly stiffer, leading to higher rms acceleration, signifies the increased transmission of road disturbance

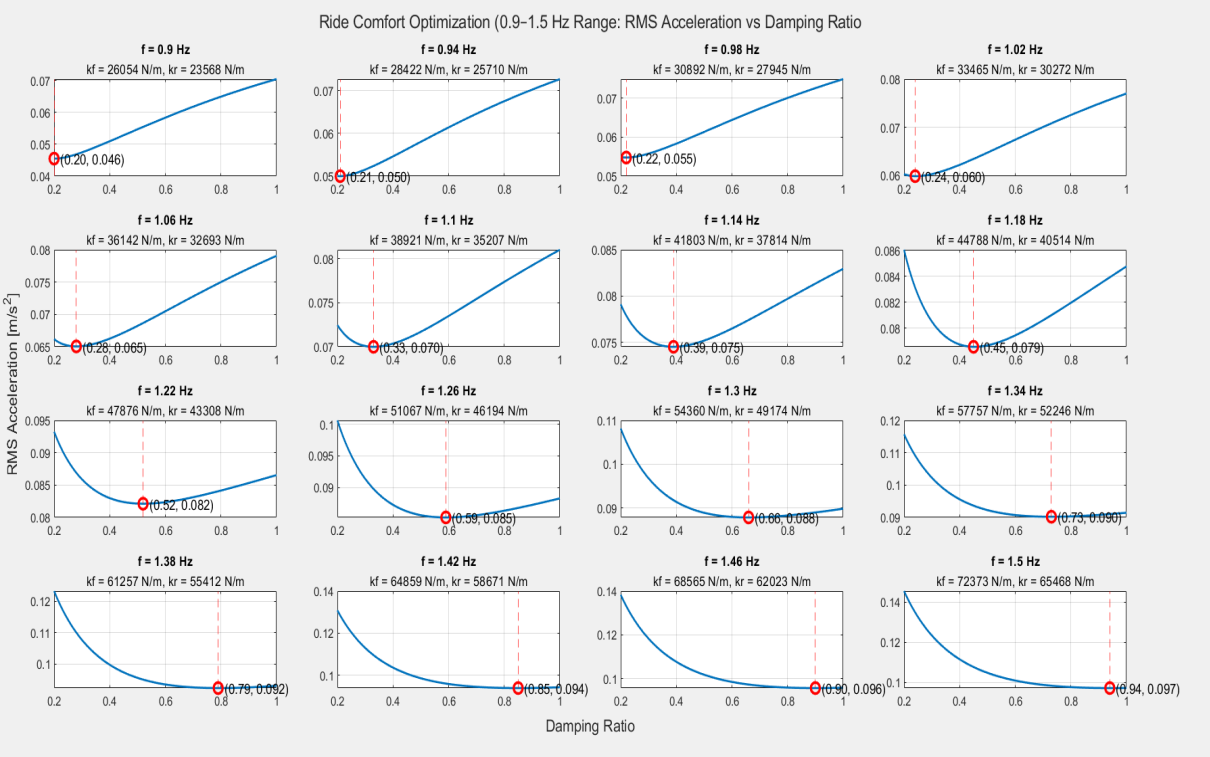


Figure 48: Ride Comfort Optimization (Narrow range frequency)

- Narrowed Frequency Range (1.14 Hz to 1.24 Hz)

  A narrower frequency range is selected around the body resonance region, where the system is most sensitive.

  Within the range:

  - RMS acceleration shows strong sensitivity to both frequency and damping
  - Small changes in frequency result noticeable changes in response

3.4.4 Improved Ride Setup

Based on the parameter study, the ride setup can be improved, since reducing the frequency below 1.2 value would move the vehicle towards a softer suspension response that is less representative of a medium passenger car. Therefore, the improvement strategy is based on damping optimisation at constant ride frequency, rather than reducing suspension stiffness further.

The results show that, at frequencies close to the selected value, RMS acceleration decreases as rear damping ratio increases, which indicates better control of the delayed rear axle response and improved suppression of bounce-pitch coupling. For example, at 1.2 Hz, increasing the rear damping ratio from $\zeta r = 0.2$ to $\zeta r = 0.4$ reduced the RMS acceleration from 0.0896 m/s² to 0.0840 m/s² for $\zeta f = 0.2$, and from 0.0883 m/s² to 0.0817 m/s² for $\zeta f = 0.25$. A similar trend is also visible

at 1.3 Hz, where RMS decreases from 0.1082 m/s² to 0.1002 m/s² when ζr increases from 0.2 to 0.4 at ζf = 0.2.

The front damping ratio also influences the response, but its effect is more sensitive. Increasing front damping generally reduces front suspension travel, but beyond a moderate level it does not provide a practical passenger-car solution, even if the numerical RMS continues to reduce. For this reason, damping ratios near ζf = 0.25 - 0.35 and ζr = 0.35 - 0.40 provide a more realistic balance for this vehicle category. In this region of the dataset, the model shows lower RMS values than the lightly damped cases, while avoiding the unrealistically high damping levels at the upper end of the sweep. For instance, at 1.2 Hz, ζf = 0.3, ζr = 0.4 gives RMS = 0.0809 m/s², and ζf = 0.35, ζr = 0.4 gives RMS = 0.0808 m/s², which are both improved relative to the lower-damped combinations.

From suspension design literature, the typical damping ratio range for passenger vehicles lies between ζ = 0.2 and 0.4 (Leo, 2023), which provides a practical balance between vibration isolation and body control. Based on the parameter study results within this realistic range, the combination of ζf ≈ 0.35 and ζr ≈ 0.4 produced the most favourable reduction in sprung-mass RMS acceleration without introducing excessive stiffness or harshness.

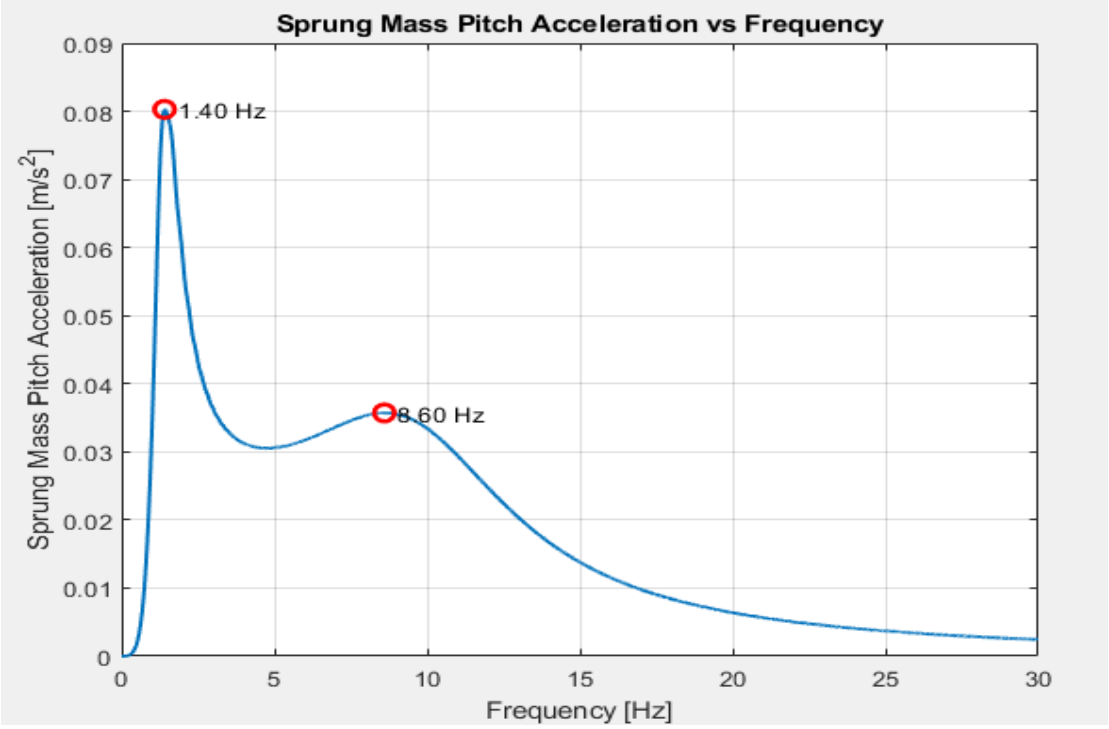


Figure 49: Unsprung Mass Vertical Acceleration vs Frequency (ff=1.2, ζf = 0.35, ζr = 0.4)

After the optimization the plot show a reduced peak amplitude when compared with figure 46, confirming that the pitch motion has been effectively controlled. This reduction directly contributes to a lower RMS acceleration, improving overall ride comfort.

Therefore, the recommended improved ride setup is to keep the ride frequency at 1.2 Hz and apply a moderate front damping ratio with slightly higher rear damping, while still maintaining the stiffness and damping characteristics expected of a standard passenger vehicle.

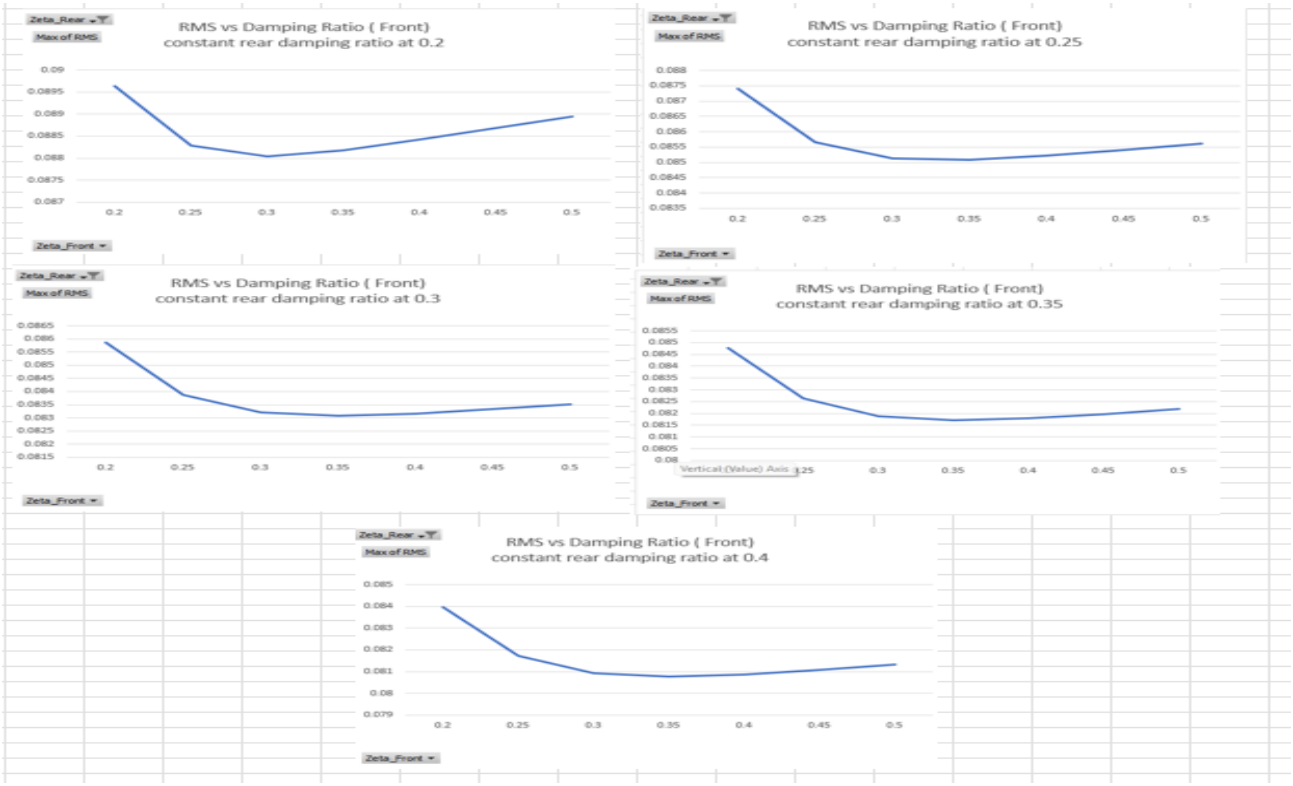


Figure 50: RMS vs Damping Ratio plots

3.5 Conclusion

The ride analysis using the 4DOF half-car model showed that the dominant body bounce frequency occurs at ~1.2–1.3 Hz, with a secondary wheel-hop mode at ~10–12 Hz. The baseline suspension setup (front stiffness 46,319 N/m, rear stiffness 41,899 N/m, damping ratio $\zeta f = 0.2$ and $\zeta r = 0.2$) resulted in a sprung mass RMS acceleration of 0.0854 m/s² at ride frequency 1.24Hz.

After optimizing the damping ratio and ride frequency RMS acceleration reduced to 0.0808 m/s² (~5.4% improvement is observed). The optimized ride frequency obtained is 1.2 Hz after parametric study.

Overall, the results confirm that reducing damping ratios directly lowers RMS acceleration, demonstrating that optimisation improves ride comfort

Appendix C: MATLAB Parameter Definition Code

```
clc;
```

%% constants

```
g=9.81;    % gravity [m/s^2]
L = 2.64;                % wheelbase (m)
%% Vehicle Parameters
M_initial = 1385;                % base vehicle mass (kg)

front = 0.6;          % 60% front
rear  = 0.4;          % 40% rear
mp = 79;                 % passenger mass (kg)
xp = 0.2;                % passenger position from front axle (m)

ml = 10;                 % luggage mass (kg)
xl = 2.3;                % luggage position from front axle (m)

W = M_initial * g;                       % base vehicle weight (N)
Wf_initial = front * W;                  % front axle load (N)
Wr_initial = rear  * W;                  % rear axle load (N)

a_initial = (Wr_initial * L) / W;              % distance from front axle to CG (m)
b_initial = L - a_initial;                     % distance from CG to rear axle (m)

M = M_initial + mp + ml;                       % total vehicle mass (kg)
W = M * g;                                  % total vehicle weight (N)

% New CG position after adding payload
a = (M_initial*a_initial + mp*xp + ml*xl) / M;      % front axle to CG (m)
b = L - a;                                  % CG to rear axle (m)

% New axle loads
Wf = W * b / L;             % front axle load (N)
```

```
Wr = W * a / L;              % rear axle load (N)


mu_total = 0.1 * M;           % total unsprung mass (kg)
musf = mu_total / 2;          % front unsprung mass (kg)
musr = mu_total / 2;          % rear unsprung mass (kg)
ms = M-musf-musr;               % Sprung mass [kg]
msf = ms*(b/L);               % front sprung mass (kg)
msr = ms*(a/L);               % rear sprung mass (kg)


ff=1.24;                    % front ride frequency [Hz]
fr=1.2*ff;                   % Rear ride frequency [Hz]
I_initial = 1728.758;         % pitch inertia (kg.m^2)

I = I_initial + mp*(xp - a)^2 + ml*(xl - a)^2;
```

%% Suspension parameters

```
kfspring = (2*pi*ff)^2*(msf);      % Front Spring stiffness [N/m]
krspring = ((2*pi*fr)^2)*(msr);      % Rear Spring stiffness [N/m]

Fof = ((msf)*g);           %Front Spring Preload [N]
For = ((msr)*g);           %Rear Spring Preload [N]

zf = 0.2;                   %front damping ratio []
zr = 0.2;                   %Rear damping ratio []

ccf = 2*sqrt(kfspring*msf);          %Front crtical damping [Ns/m]
ccr = 2*sqrt(krspring*msr);          %Rear crtical damping [Ns/m]

cfdamper = zf*ccf;                %Front Damping coefficient [Ns/m]
crdamper = zr*ccr;                %Rear Damping coefficient [Ns/m]
```

```
ktyre = 290000;       %Tyre stiffness [N/m]

zOftyre = (((msf)+musf)*g/ktyre); %Deflection of the front tyre at the equilbrium [m]
zOrtyre = (((msr)+musr)*g/ktyre); %Deflection of the rear tyre at the equilbrium [m]
%% Input values
v = 4;       % vehicle speed (m/s)

h = 0.03;      % bump height
l = 2;         % bump length

T  = l / v;   % bump duration
time = L/v;
t0 = 0.5;     % front wheel bump start time
t1 = t0 + time;% rear wheel bump start time

%% simulation run for half sinebump
load_system("DOF4_sinebump");
out = sim("DOF4_sinebump");
```

%% Approximate front and rear wheel hop frequencies

```
ridefreq_f=sqrt(kfspring/msf)/(2*pi);
ridefreq_r=sqrt(krspring/msr)/(2*pi);
fhf=sqrt((ktyre+kfspring)/musf)/(2*pi);
fhr=sqrt((ktyre+krspring)/musr)/(2*pi);
ridefreq_f
ridefreq_r
fhf
fhr

rms_sprung=rms(out.zsdotdot.data)
% rms_unsprungf=rms(out.zusfdotdot.data)
% rms_unsprungr=rms(out.zusrdotdot.data
```

%% Road input front

```
 figure;
```

```
plot(out.zrf.time,out.zrf.data,'Linewidth',1.5);
hold on;
plot(out.zrr.time,out.zrr.data,'Linewidth',1.5);
xlabel('Time [s]');
ylabel('Road Input Front [m]');
legend('Front','Rear')
grid on
xlim([0,5]);
title('Road Input vs Time');
```

%% masses vertical Displacement

```
figure;
xlabel('Time [s]');
ylabel('Sprung Mass Vertical Displacement [m]');
xlim([0 5]);
grid on
title('Mass Vertical Displacement vs Time');
hold on
plot(out.zs.time,out.zs.data,'Linewidth',1.5);
plot(out.zusf.time,out.zusf.data,'Linewidth',1.5);
plot(out.zusr.time,out.zusr.data,'Linewidth',1.5);
plot(out.tout,out.zrf.data,'Linewidth',1.5);
plot(out.tout,out.zrr.data,'Linewidth',1.5);

legend('Sprung','Front Unsprung','Rear Unsprung','Front Road Input','Rear Road Input');
hold off
```

%% mass vertical velocity

```
figure;
plot(out.zsdot.time,out.zsdot.data,'Linewidth',1.5);
hold on
plot(out.zusfdot.time,out.zusfdot.data,'Linewidth',1.5);
plot(out.zusrdot.time,out.zusrdot.data,'Linewidth',1.5);
xlabel('Time [s]');
```

```
ylabel('Mass Vertical velocity [m/s]');
grid on
title('Masses Vertical velocity vs Time');
legend('Sprung','Front Unsprung','Rear Unsprung');
hold off;
xlim([0 5]);
```

%% mass vertical acc

```
figure;
plot(out.zsdotdot.time,out.zsdotdot.data,'Linewidth',1.5);
hold on
plot(out.zusfdotdot.time,out.zusfdotdot.data,'Linewidth',1.5);
plot(out.zusrdotdot.time,out.zusrdotdot.data,'Linewidth',1.5);
xlabel('Time [s]');
ylabel('Mass Vertical acc [m/s^2]');
grid on
title('Masses Vertical Acc vs Time');
legend('Sprung','Front Unsprung','Rear Unsprung');
hold off;
xlim([0 5]);
```

%% simulation run for frequency analysis

```
load_system("DOF4_frequency_input.slx")
%open_system("DOF4_frequency_input.slx")
out = sim("DOF4_frequency_input.slx");
%% Sprung mass vertical acc
dt=0.001;
figure;
[amps, phases, freqs] = fft_V(out.zsdotdot.data, dt);
plot(freqs, amps, 'LineWidth', 1.5);
hold on;
valid_idx = freqs > 0.5;
amps_f = amps(valid_idx);
freqs_f = freqs(valid_idx);
[pks, locs] = findpeaks(amps_f, freqs_f);
```

```
[pks_sorted, sort_idx] = sort(pks, 'descend');
top2_pks = pks_sorted(1:2);
top2_freqs = locs(sort_idx(1:2));
plot(top2_freqs, top2_pks, 'ro', 'MarkerSize', 8, 'LineWidth', 2);
text(top2_freqs(1), top2_pks(1), sprintf('  %.2f Hz', top2_freqs(1)));
text(top2_freqs(2), top2_pks(2), sprintf('  %.2f Hz', top2_freqs(2)));
xlabel('Frequency [Hz]');
ylabel('Sprung Mass Vertical Acceleration [m/s^2]');
title('Sprung Mass Vertical Acceleration vs Frequency');

grid on;
xlim([0 30]);
```

%% UnSprung mass vertical acc

```
figure;
[ampusf,phaseusf,frequsf]=fft_V(out.zusfdotdot.data,dt);
[ampusr,phaseusr,frequsr]=fft_V(out.zusrdotdot.data,dt);
plot(frequsf,ampusf,'Linewidth',1.5);
xlabel('Frequency [Hz]');
ylabel('UnSprung Mass Vertical Acceleration [m/s^2]');
xlim([0 30]);
grid on
title('UnSprung Mass Vertical Acceleration vs Frequency');
hold on
plot(frequsr,ampusr,'Linewidth',1.5);
legend('Front','Rear');
hold off
```

%% Sprung mass pitch acc

```
figure;
[ampps, phaseps, freqps] = fft_V(out.pitchdotdot.data, dt);
plot(freqps, ampps, 'LineWidth', 1.5);
hold on;
valid_idx = freqs > 0.5;
amps_f = ampps(valid_idx);
freqs_f = freqps(valid_idx);
```

```
[pks, locs] = findpeaks(amps_f, freqs_f);
[pks_sorted, sort_idx] = sort(pks, 'descend');
top2_pks = pks_sorted(1:2);
top2_freqs = locs(sort_idx(1:2));
plot(top2_freqs, top2_pks, 'ro', 'MarkerSize', 8, 'LineWidth', 2);
text(top2_freqs(1), top2_pks(1), sprintf('  %.2f Hz', top2_freqs(1)));
text(top2_freqs(2), top2_pks(2), sprintf('  %.2f Hz', top2_freqs(2)));
xlabel('Frequency [Hz]');
ylabel('Sprung Mass Pitch Acceleration [m/s^2]');
title('Sprung Mass Pitch Acceleration vs Frequency');

grid on;
xlim([0 30]);
```

Appendix D: Plots

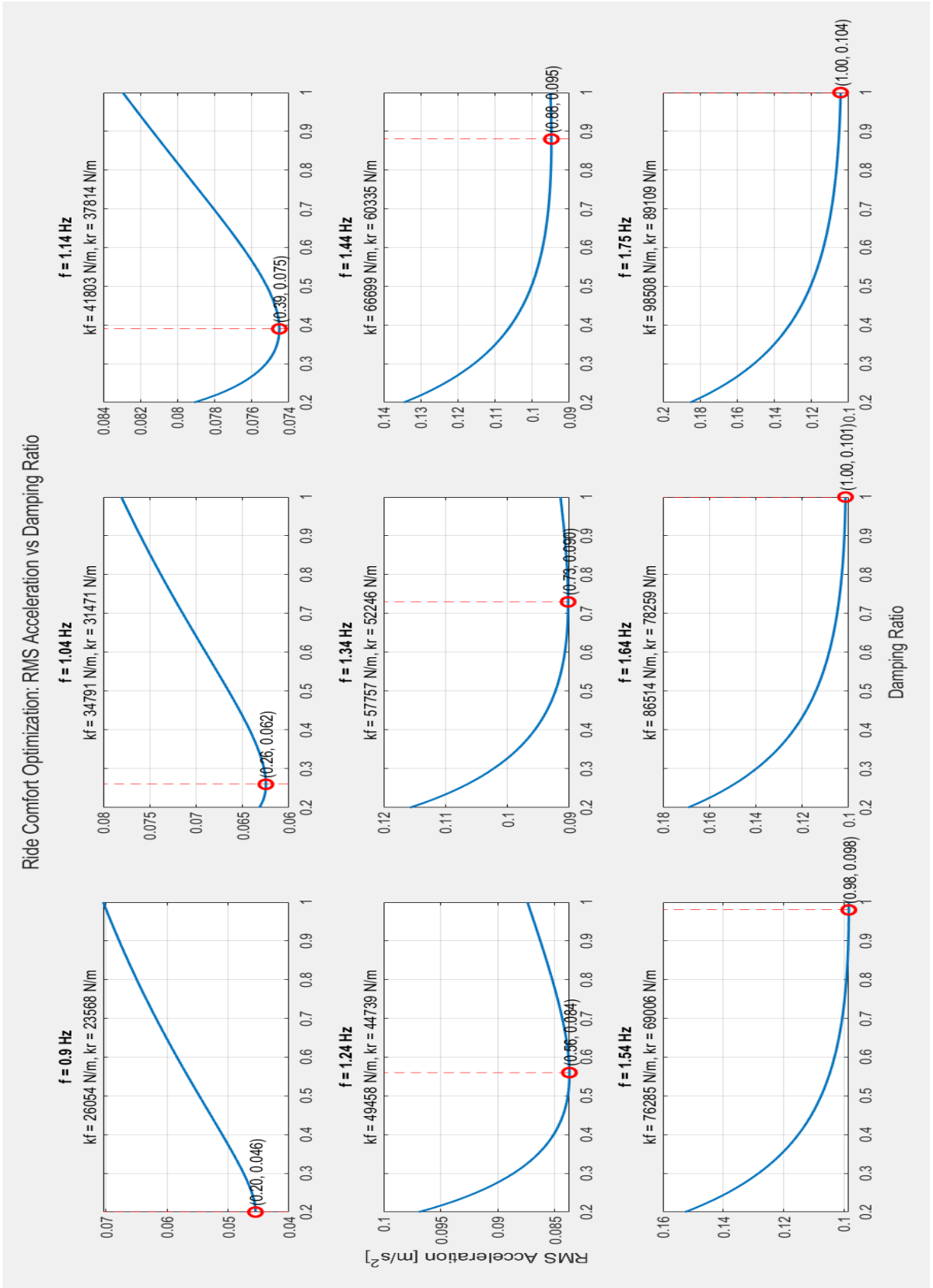


Figure 47: Wide Frequency Range (0.9 Hz to 1.75 Hz)

Note: Figure file is also uploaded in MATLAB file (wide range freq.fig)

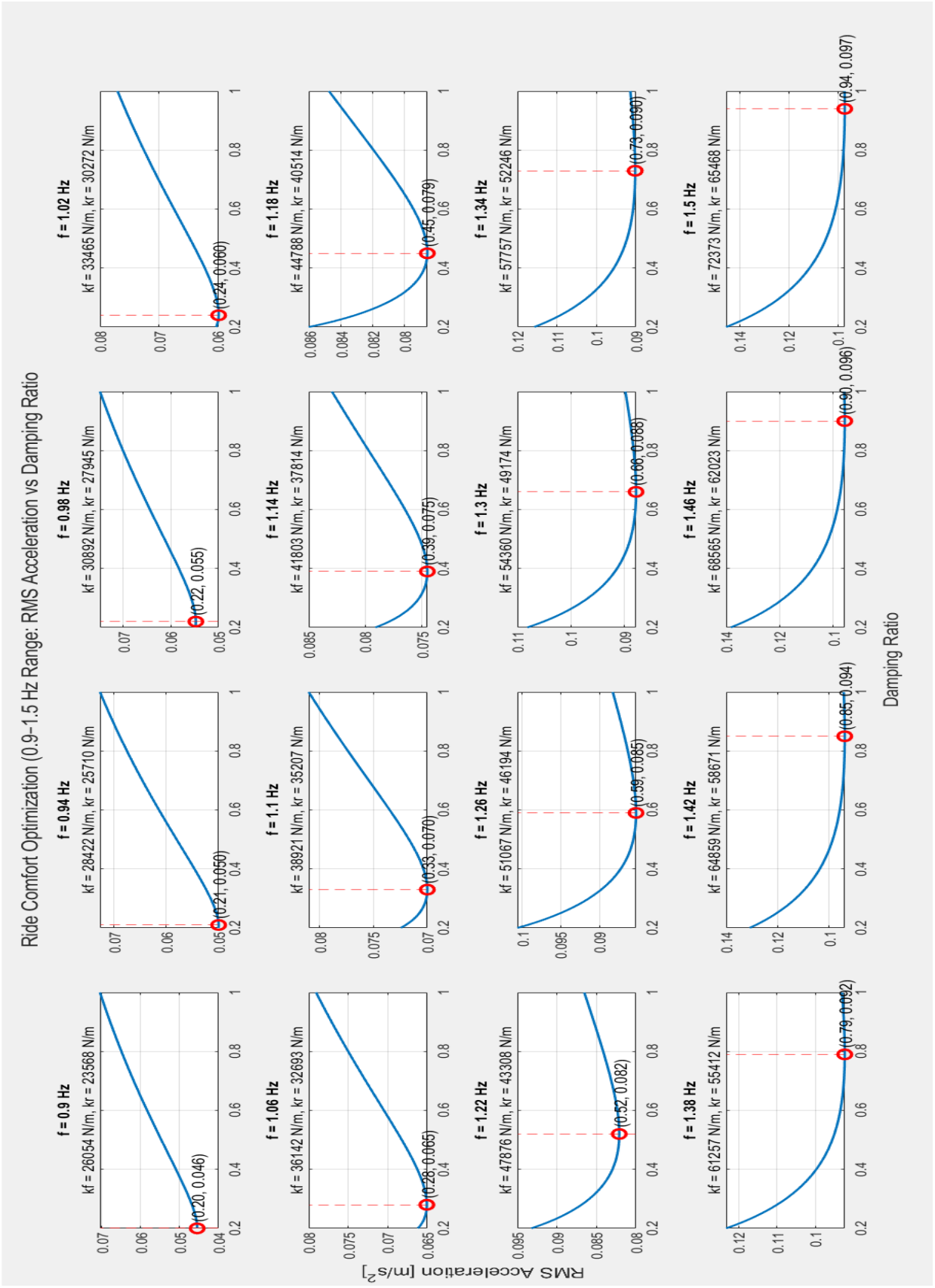


Figure 48: Ride Comfort Optimization (Narrow range frequency)

Note: Figure file is also uploaded in MATLAB file (Narrow range freq.fig)

Appendix E: Simulink Model Implementation

Figure 51: Half sinewave Input Model

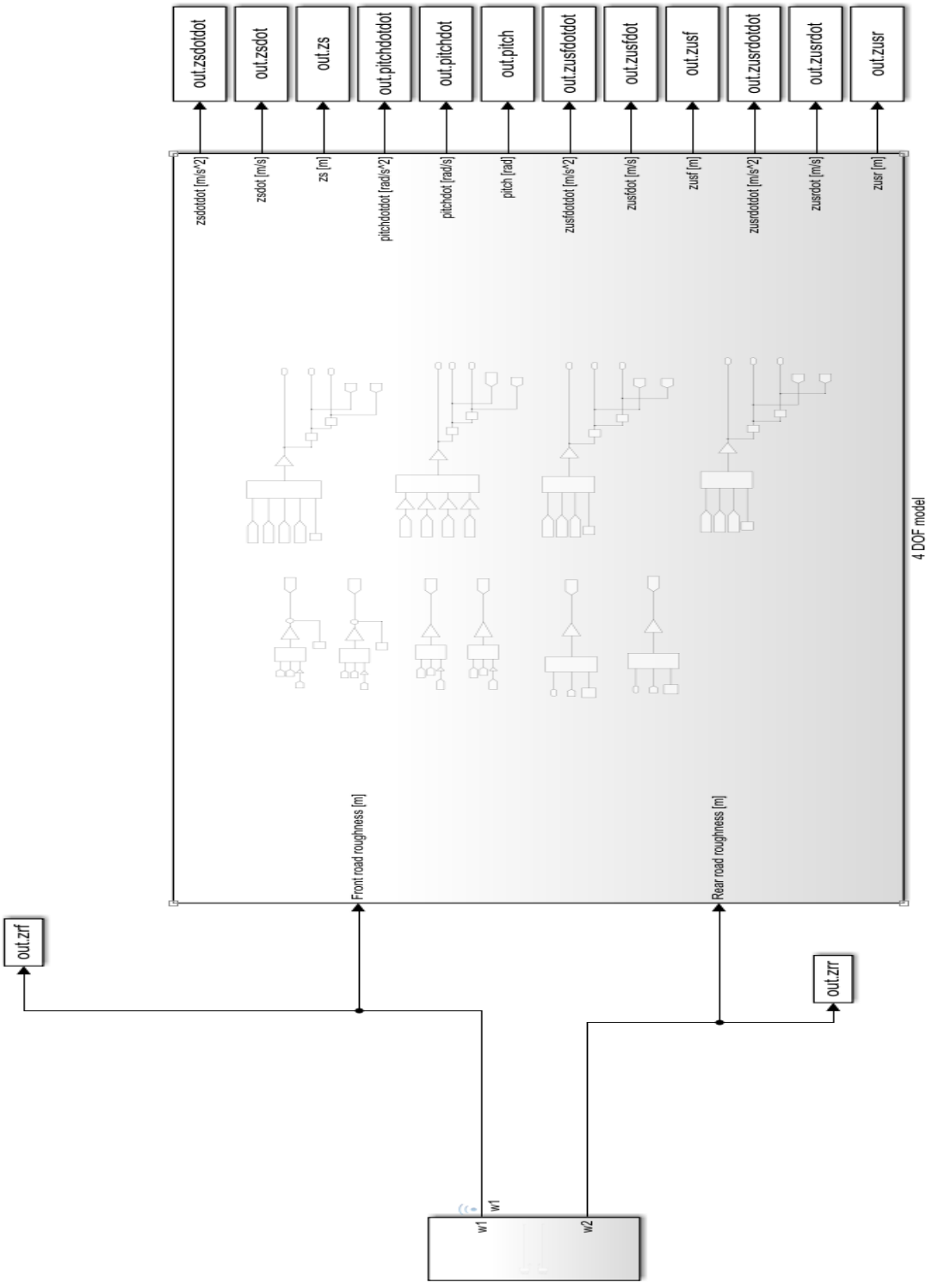
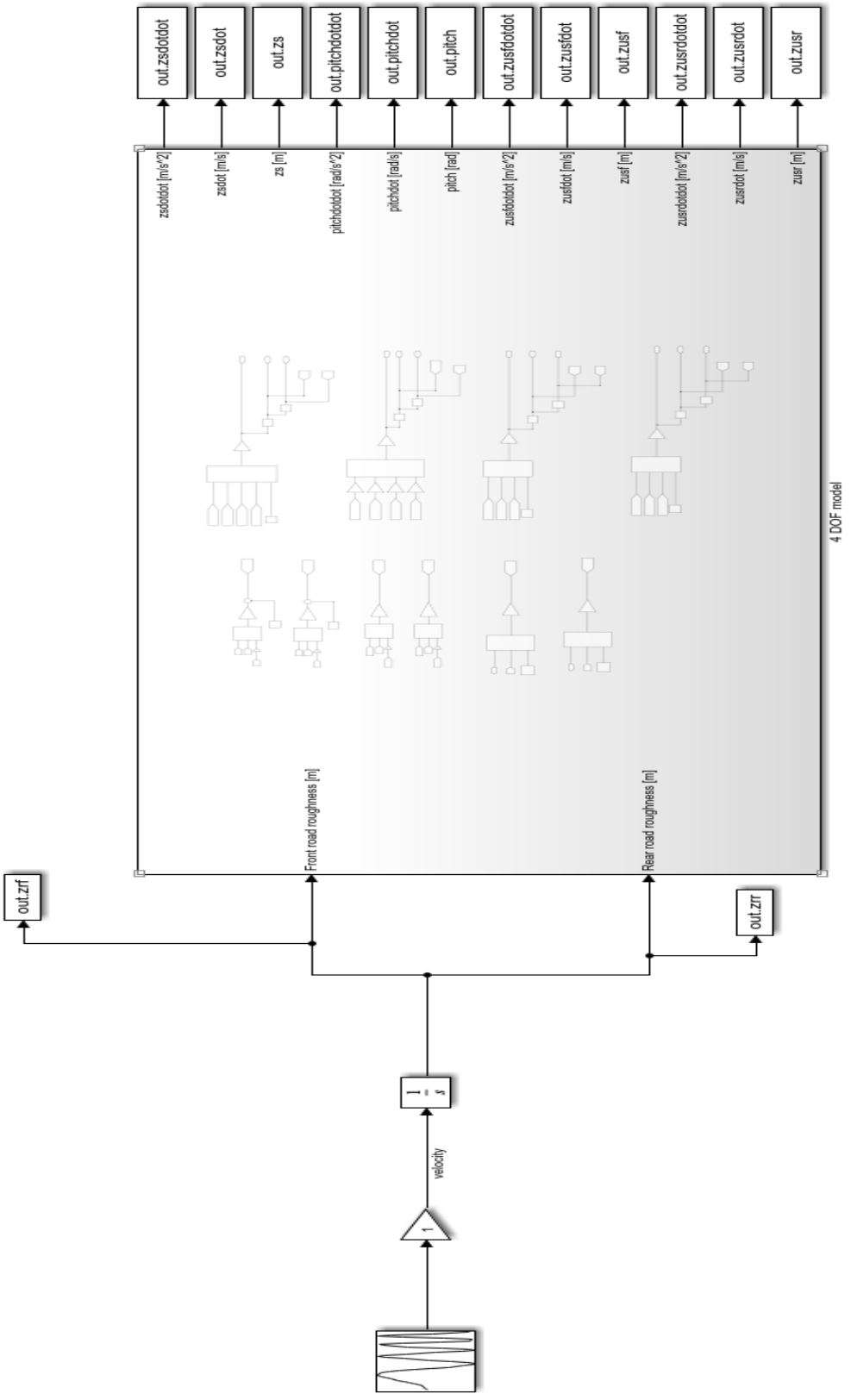


Figure 52: Frequency Input Model



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